
LC Classical Mechanics and Relativity 2 Lecture Notes

Year 1 Semester 2

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Lecture 1 - Mechanics I: Rigid Bodies

Recommended Course Text

Introduction to Classical Mechanics, Oxford University Press

1 Rotational Mechanics

We've been introduced to a number of equations for translational mechanics such as:

$$F = ma$$

$$W = \int \vec{F} \cdot d\vec{r}$$

$$K = \frac{1}{2}mv^2$$

In simple cases that we'll deal with on this course, these all have equivalent formulae for rotational mechanics. Instead of position r , we have angle θ . Instead of velocity v , we have angular velocity ω . Instead of acceleration a , we have angular acceleration α . Instead of mass m , we have moment of inertia I . Instead of force F , we have torque τ . Etc...

The main goal is to discover alternative formulae for rotational quantities, and to understand how they relate to the translational ones.

2 Centre of Mass

We can consider two possible cases:

1. A single point like object, such as a particle.
2. An extended object, such as a rod or a disc, made up of many particles.

Given a system of i discrete masses, m_i , located at positions $\vec{r}_i$, the centre of mass $\vec{R}$ is given by:

$$M\vec{R} = \sum_i m_i \vec{r}_i$$

Where $M = \sum_i m_i$ is the total mass. For every component mass m_i , we consider two types of net forces acting upon it:

- Internal forces, $\vec{f}_i^{\text{int}}$.
- External forces, $\vec{f}_i^{\text{ext}}$.

The internal forces are those exerted by other masses in the system, and the external forces are those exerted by objects outside the system. Hence:

$$\vec{f}_i^{\text{int}} = \sum_{j \neq i} \vec{f}_{ij}$$

We want to consider how an object moves, and if we assume the object has a constant mass and is rigid (does not deform) then we can consider the motion of the centre of mass.

$$\begin{aligned}
 M\ddot{\vec{R}} &= \sum_i m_i \ddot{\vec{r}}_i = \sum_i m_i \vec{a}_i = \sum_i (\vec{f}_i^{\text{int}} + \vec{f}_i^{\text{ext}}) \\
 &= \sum_i \left(\sum_{j \neq i} \vec{f}_{ij} + \vec{f}_i^{\text{ext}} \right) \\
 &= \sum_i \sum_{j \neq i} \vec{f}_{ij} + \sum_i \vec{f}_i^{\text{ext}}
 \end{aligned}$$

However, if one internal mass exerts a force on another, then the second mass exerts an equal and opposite force on the first. Hence, the internal forces all nicely cancel out:

$$\sum_i \sum_{j \neq i} \vec{f}_{ij} = 0$$

$$M\ddot{\vec{R}} = \sum_i \vec{f}_i^{\text{ext}} = \vec{F}^{\text{ext}}$$

This is the analogue of Newton's 2nd Law but for a non-point-like object. For a rigid body, the centre of mass moves as if the entirety of the object's mass were concentrated at that point and all external forces were applied at that point.

For a rigid body, we can break down the motion into two components:

- Translational motion of the centre of mass.
- Rotational motion about the centre of mass (in the centre of mass frame)

3 Continuous Objects

However, many extended objects can be thought of as having a continuous distribution of mass, rather than being made up of finite discrete masses.

We define the density $\rho(\vec{r})$, such that $\rho(\vec{r})dV$ is the mass contained in an infinitesimal volume element dV around the point $\vec{r}$. While a point is massless, a small volume can contain a mass, albeit a tiny mass element. We can therefore get the total mass by integrating across all of these mass elements (effectively, we just swap sums for integrals).

Note that:

$$dV = dx dy dz \equiv d^3\vec{r}$$

Thu 22 Jan 2026 15:00

Lecture 2 - Mechanics II: Continuous Mass and Angular Momentum

1 Updated CoM for Continuous Mass

We can update:

$$M = \sum_i m_i \quad \rightarrow \quad M = \int dm = \int \rho(\vec{r}) dV$$

Hence our centre of mass becomes:

$$M\vec{R} = \sum_i \vec{r}_i m_i \quad \rightarrow \quad M\vec{R} = \int \vec{r} dm = \int \vec{r} \rho(\vec{r}) dV$$

This point is effectively the point at which you could hold an object and it would balance with no motion. It does not always lie within the object itself however, i.e. for a doughnut, the centre of mass lies within the middle of the hole.

We can also generally use symmetry, as the CoM lies on the axes (or planes) of symmetry. For example, for a homogeneous sphere, the CoM lies at the centre of the sphere, as infinitely many planes of symmetry intersect at this point.

2 Example: Inhomogeneous Rod

Consider a cylindrical rod of length L and cross-sectional area A with a density that varies across its length as $\rho(x) = Cx^2$. The axis of the rod is along the x -axis, with faces at $x = 0, x = L$

We immediately know that the CoM lies on the x -axis by symmetry.

Our mass elements are infinitesimal slices from x to $x + dx$. We have a volume element:

$$dv = A dx$$

and hence a mass element:

$$dm = \rho dv = C A x^2 dx$$

Therefore:

$$M = \int_0^L A C x^2 dx = A C \left[\frac{x^3}{3} \right]_0^L = \frac{A C L^3}{3}$$

And to get the CoM:

$$M R = \int_0^L x dm = \int_0^L x C A x^2 dx = A C \int_0^L x^3 dx = A C \left[\frac{x^4}{4} \right]_0^L = \frac{A C L^4}{4}$$

Hence the centre of mass is:

$$R = \frac{M R}{M} = \frac{A C L^4 / 4}{A C L^3 / 3} = \frac{3L}{4}$$

3 Example: Homogeneous Hemisphere

Consider a homogeneous hemisphere of radius R with its flat face on the xy -plane and the centre point of the hypothetical full sphere at the origin.

By symmetry, the CoM lies on the z -axis. We can take a mass element as a thin disc of radius r and thickness dz at height z . The mass of this disc is:

$$dm = \rho \pi r^2 dz$$

However we need to know r as a function of z to be able to integrate. By Pythag. we have:

$$r^2 + z^2 = R^2 \quad \rightarrow \quad r^2 = R^2 - z^2$$

Hence:

$$dm = \rho \pi (R^2 - z^2) dz$$

$$\Rightarrow M = \int dm = \int_0^R \rho \pi (R^2 - z^2) dz = \rho \pi \left[R^2 z - \frac{z^3}{3} \right]_0^R = \frac{2}{3} \rho \pi R^3$$

We now have:

$$M z_{\text{COM}} = \int_0^R z dm = \int_0^R z \rho \pi (R^2 - z^2) dz = \rho \pi \left[\frac{R^2 z^2}{2} - \frac{z^4}{4} \right]_0^R = \frac{1}{4} \rho \pi R^4$$

Hence:

$$z_{\text{COM}} = \frac{M z_{\text{COM}}}{M} = \frac{1/4 \rho \pi R^4}{2/3 \rho \pi R^3} = \frac{3R}{8}$$

4 Angular Momentum

Consider an object with momentum $\vec{p}$ moving relative to an origin O , at distance $\vec{r}$. This object is defined to have angular momentum relative to the origin of:

$$\vec{L}(0) = \vec{r} \times \vec{p} = \vec{r} \wedge \vec{p}$$

For an extended system, we have:

$$\vec{L}(0) = \sum_i \vec{r}_i \times \vec{p}_i$$

Noting that any applied force will change the angular momentum. This becomes reasonably simple if we assume that the force is applied such that only the magnitude of the momentum changes, preserving the orientation.

We therefore restrict the rotational motion to be parallel to a plane, i.e. the xy -plane. In this case, the axis of rotational motion is parallel to the z -axis.

For example, consider an object rotating in the xy -plane with tangential velocity v . The axis of rotation is the z -axis. The object makes varying angle θ with the x -axis.

The object makes some $d\theta$, and rotational dl . We hence have:

$$\vec{v} = \frac{d\vec{l}}{dt}$$

Or:

$$v = \frac{dl}{dt} = \frac{r d\theta}{dt} = r \dot{\theta}$$

We define the angular velocity ω as $\omega = \dot{\theta}$.

Thu 29 Jan 2026 15:00

Lecture 3 - Mechanics III: Moment of Inertia

1 Moments of Inertia Examples

1.1 Point Mass on a Light Ring

Consider a point mass on a light ring. The ring has radius R and the point masses axis of rotation runs through the ring centre.

This is trivially:

$$I = mR^2$$

1.2 Uniform Ring

Consider the same example, but instead of a point mass the ring is massive, with mass M . All of the mass still sits at the same distance from the axis of rotation.

Hence it is still $I = MR^2$.

2 Uniform Solid Disk

Consider a uniform solid disk of radius R and constant mass density ρ . Similar to CoM problems we need to break it down and consider the object as being a sum over multiple elements, i .

$$I = \sum_i m_i r_i^2$$

As the size of each element becomes infinitesimal, this becomes an integral:

$$I = \int dI = \int r^2 dm$$

To make this integral easier we want to think of elements at a fixed r , i.e. considering infinitesimal rings.

3 Thick Solid Disk

Consider a disk now being 3D, with height h . The mass element for each infinitesimal ring is now:

$$dm = \rho dV = \rho h (\pi(r + dr)^2 - \pi r^2)$$

Expanding and disregarding the dr^2 term:

$$= \rho h 2\pi r dr$$

Hence for the ring, the contribution to the moment of inertia is:

$$dI = r^2 dm = 2\pi \rho h r^3 dr$$

And integrating:

$$I = \int_0^R dI = \int_0^R 2\pi \rho h r^3 dr$$

$$= 2\pi\rho h \left[\frac{r^4}{4} \right]_0^R = \frac{\pi\rho h R^3}{2}$$

And to swap from density into mass we use $M = \rho V$:

$$\rho = \frac{M}{V} = \frac{M}{\pi R^2 h}$$

Hence:

$$I = \frac{\pi h R^4}{2} \times \frac{M}{\pi R^2 h} = \frac{MR^2}{2}$$

4 Parallel Axis Theorem

So far, we assume that the axis of rotation is collinear with the centre of mass. However, we may care about the moment of inertia passing through an axis which does not pass through the CoM (or indeed the object itself). Rather than doing it manually, we can make a shortcut if we know the result for the moment of inertia through a parallel axis through the CoM.

Consider an arbitrary shaped object with axis A_0 through the CoM, I_0 . Consider a parallel axis A outside of the object, I .

The distance between the axis is given as R . We select a coordinate system with the origin at the CoM.

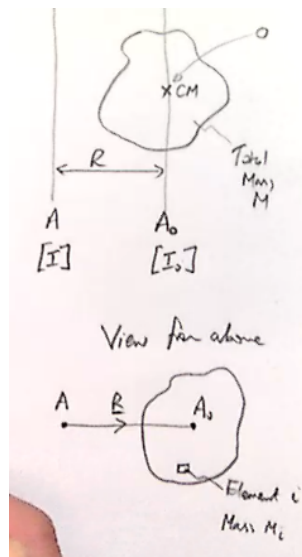


Figure 3.1

Define r_i as the vector between A (at the same vertical position) and a random point in the object, and r_i^0 being the vector from A_0 to the same point.

$$I_A = \sum_i m_i r_i^2$$

By vector addition:

$$\begin{aligned} I_A &= \sum_i m_i (\vec{R} + \vec{r}_i^0)^2 \\ &= \sum_i m_i (R^2 + r_i^{02} + 2\vec{R} \cdot \vec{r}_i^0) \end{aligned}$$

R is constant, hence:

$$\begin{aligned} I_A &= R^2 \sum_i m_i + \sum_i m_i r_i^{02} + 2\vec{R} \cdot \sum_i m_i \vec{r}_i^0 \\ &= R^2 M + I_0 + 2\vec{R} \cdot \sum_i m_i (\vec{r}_i^0)^2 \end{aligned}$$

However we define the centre of mass with (noting we defined the origin at the CoM):

$$r_{\text{CM}} = 0 = \frac{\sigma_i m_i r_i^0}{M}$$

So:

$$\sigma_i m_i (\vec{r}_i^0)^2 = 0$$

Hence finally:

$$I_A = I_0 + MR^2$$

Tue 03 Feb 2026 11:00

Lecture 4 - Mechanics IV: Perp Axes, Radius of Gyration and Angular Momenta

1 Perpendicular Axis Theorem

This only applies in flat 2D objects, or objects which can be approximated as them, for example a pancake or a coin. Suppose the object sits in the XY plane. The perpendicular axis theorem says that the moment of inertia about the z-axis is:

$$I_z = I_x + I_y$$

For rotation in the x-axis:

$$I_x = \sum_i m_i y_i^2$$

And in the y:

$$I_y = \sum_i m_i x_i^2$$

By Pythag:

$$\begin{aligned} r_i^2 &= x_i^2 + y_i^2 \\ \Rightarrow \sum_i m_i x_i^2 + \sum_i m_i y_i^2 &= \sum_i m_i r_i^2 \\ I_y + I_x &= I_z \end{aligned}$$

1.1 Spinning a Coin

Consider spinning a coin vertically on a table. Suppose the coin spins about the y-axis (vertical) and the z-axis is the axis of symmetry.

By symmetry, $I_x = I_y = I_{\text{end}}$. By the perpendicular axes theorem, $I_x + I_y = 2I_{\text{end}} = I_z$.

Hence:

$$2I_{\text{end}} = I_z = \frac{MR^2}{2} \Rightarrow I_{\text{end}} = \frac{MR^2}{4}$$

2 Radius of Gyration

The radius of gyration, K is the effective radius at which we could locate all of the mass of an extended body to get the same moment of inertia about an axis passing through the CoM.

It is defined by:

$$I_0 = MK^2 \Rightarrow K = \sqrt{\frac{I_0}{M}}$$

We know that $I_z = \frac{1}{2}MR^2$ and by the definition of the radius of gyration:

$$I_z = MK^2$$

Hence:

$$\frac{1}{2}MR^2 = MK^2 \Rightarrow K^2 = R^2/2$$

3 Angular Momentum

We have defined angular momentum as:

$$\vec{L} = \vec{r} \wedge \vec{p}$$

Angular momentum, just like linear momentum, is conserved. "Conserved" means that it is not time dependant, i.e.:

$$\frac{d\vec{L}}{dt} = 0$$

$$\begin{aligned} \frac{d\vec{L}}{dt} &= \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt} \\ &= \vec{v} \times \vec{p} + \vec{r} \times \vec{F} \\ &= \vec{v} \times m\vec{v} + \vec{r} \times \vec{F} \\ &= 0 + \vec{r} \times \vec{F} \end{aligned}$$

This second term is torque. There is always a centripetal force for circular motion, so $|\vec{F}| = 0$ does not hold. However, as $\vec{r}$ and $\vec{F}$ are in the same direction, this term does become zero. Accordingly, angular momentum is conserved under radially inwards (central) forces.

And since $\vec{r}$ is defined around a specific origin point, angular momentum is defined about a point in space and varies from point to point.

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Lecture 5 - Mechanics V: Rigid Pendula

Reminder, angular momentum:

- A vector, perpendicular to $\vec{r}, \vec{p}$, given by $\vec{L} = \vec{r} \wedge \vec{p}$.
- Conserved under central forces (i.e. forces pointing radially).
- Must be defined about some origin, hence is defined about a point in space.

Reminder, torque:

- A vector, perpendicular to $\vec{r}, \vec{F}$, given by $\vec{\tau} = \vec{r} \wedge \vec{F}$.
- Also defined about a point.
- For a fixed force, the torque grows with distance away from the fulcrum.
- Can also be approached using the magnitude $\tau = I\ddot{\theta} = I\dot{\omega}$

1 Example: Small Oscillations of a Simple Pendulum

Consider a mass at a frictionless pivot (the origin). The bob has mass m and makes angle θ with the horizontal. The rod connecting them is light.

Considering the torque about O :

$$\vec{\tau} = \vec{r} \wedge \vec{F} = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ l \sin \theta & -l \cos \theta & 0 \\ 0 & -mg & 0 \end{bmatrix} = \hat{k}(-mgl \sin \theta)$$

We also have that:

$$\tau = I\ddot{\theta}$$

Hence:

$$I\ddot{\theta} = -mgl \sin \theta \quad (1)$$

For a point mass, the moment of inertia at distance $R = l$ is $I = ml^2$. As we have small oscillations, $\sin \theta \approx \theta$. Hence:

$$ml^2\ddot{\theta} \approx -mgl\theta$$

$$\ddot{\theta} = -\frac{g\theta}{l}$$

This gives us simple harmonic motion with:

$$\omega^2 = \frac{g}{l}$$

where:

$$\theta = \theta_0 \cos \omega t$$

Given that simply harmonic motion is defined by:

$$\ddot{x} + \omega^2 x = 0$$

$$\text{Hence } T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{g}}.$$

2 Rigid Pendulum

Consider a pendulum of any arbitrary shape and mass distribution where the pivot point is the top edge of the object.

Suppose the vector from the pivot point to the centre of mass (with downward force mg) has length l and is at angle from the downward vertical θ .

We still have:

$$\vec{\tau} = \vec{r} \wedge \vec{F} = \vec{l} \times mg = -lmg \sin \theta$$

$$\tau = I_A \ddot{\theta}$$

Hence:

$$I_A \ddot{\theta} = -lmg \sin \theta \approx -lmg \theta$$

Hence again:

$$\ddot{\theta} = \frac{-mgl\theta}{I_A}$$

This still gives us simple harmonic motion, but our ω^2 is now $\omega^2 = mgl/I_A$. If we know the moment of inertia about the CoM, we can use I_0 with the parallel axes theorem hence:

$$\omega^2 = \frac{mgl}{I_0 + ml^2}$$

We can also invoke the radius of gyration to make it mass independent and depend only on the shape (which is encoded into K):

$$\omega^2 = \frac{mgl}{mK^2 + ml^2}$$

$$= \frac{gl}{K^2 + l^2}$$

The time period is again:

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{K^2 + l^2}{lg}}$$

Lets check that for the simple pendulum where the shape is a point mass, $I_0 = 0 \implies k = 0$.

Hence applying the general formula to a point mass case gives:

$$T = 2\pi \sqrt{l^2/gl} = 2\pi \sqrt{\frac{l}{g}}$$

Which is as-expected!

3 Example: Solid Uniform Rod

Suppose our object is a solid uniform rod of mass M oscillating with angle θ and of length d . The CoM lies at O at position $d/2$ on the rod relative to the axis A at the pivot. Does this oscillate faster or slower than the point mass example?

It can be shown that the moment of inertia about the CoM is:

$$I_0 = Md^2/12$$

Using the parallel axes theorem:

$$I_A = I_0 + \left(\frac{d}{2}\right)^2 M = \frac{Md^2}{12} + \frac{Md^2}{4} = \frac{Md^2}{3}$$

We have shown that:

$$\omega = \sqrt{\frac{mgl}{I_A}}$$

Where (here) $l = d/2$.

$$\omega = \sqrt{\frac{3mgd}{2md^2}} = \frac{3g}{2d}$$

This gives:

$$T = 2\pi\sqrt{\frac{2}{3}\frac{d}{g}}$$

Which is a slower oscillation compared to a simple pendulum.

4 Massive Pulleys

Consider a pulley system with a massless string pulled over a uniform pulley (at a fixed support) with radius R . The LHS dangling mass has mass m_1 and the RHS has a smaller mass m_2 . The LHS has tension T_1 and RHS tension T_2 .

Equations of Motion for the Masses

Using $F = ma$ on the two masses gives:

$$m_1g - T_1 = m_1a \quad (1)$$

$$T_2 - m_2g = m_2a \quad (2)$$

Case 1: Massless Pulley

If the pulley is massless, the two tensions are equal $T_1 = T_2 = T$

$$m_1g - T = m_1a \quad (1b)$$

$$T - m_2g = m_2a \quad (2b)$$

$1b + 2b$ gives:

$$\begin{aligned} m_1g - m_2g &= (m_1 + m_2)a \\ \Rightarrow a &= \left(\frac{m_1 - m_2}{m_1 + m_2}\right)g \end{aligned}$$

Case 2: Massive Pulley The pulley now has mass M . In order to get the pulley moving, there needs to be some kind of net torque on the pulley. This means that the two tensions T_1 and T_2 can no longer be assumed to be equal.

$$\begin{aligned} \tau &= I\ddot{\theta} \equiv I\alpha \\ \vec{\tau} &= \vec{r} \wedge \vec{F} = \vec{R} \wedge (T_2 - T_1) \end{aligned}$$

Hence:

$$I\alpha = R(T_2 - T_1) \quad (3)$$

And as its a uniform disk:

$$I = \frac{MR^2}{2}$$

And converting from angular formulae to linear formulae:

$$v = R\omega \Rightarrow a = R\alpha$$

Inserting these into (3):

$$\begin{aligned} \frac{MR^2}{2} \frac{a}{R} &= R(T_2 - T_1) \\ \Rightarrow \frac{Ma}{2} &= T_2 - T_1 \end{aligned} \quad (4)$$

And inserting our general equations of motion:

$$T_2 = m_2g - m_2a, \quad T_1 = m_1g + m_1a$$

$$\frac{Ma}{2} = (m_2g - m_2a) - (m_1g + m_1a)$$

$$\frac{Ma}{2} = (m_2 - m_1)g - (m_1 + m_2)a$$

$$\frac{Ma}{2} + (m_1 + m_2)a = (m_2 - m_1)g$$

$$a\left(\frac{M}{2} + m_1 + m_2\right) = (m_2 - m_1)g$$

$$a = \frac{(m_2 - m_1)g}{\frac{M}{2} + m_1 + m_2}$$

Tue 10 Feb 2026 11:00

Lecture 6 - Mechanics VI: Vectors, Moving Axes and Gyroscopes

1 Angular Velocity and Momentum as Vectors

We define the *angular velocity vector*:

$$\vec{\omega} = \omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}$$

For an object which is rotating in a single plane, the angular velocity vector is perpendicular to the plane, i.e. pointing parallel to the axis of rotation. It's direction can be given by the Right Hand Rule.

We can likewise define the angular momentum vector:

$$\vec{L} = I\vec{\omega}$$

Where $L_x = I_x \omega_x$ etc. As the moment of inertia is different in each axis (often), $\vec{L}$ and $\vec{\omega}$ may not be parallel.

2 Torque and Changes in Angular Momentum Direction

Torque and angular momentum are fundamentally connected via:

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

If τ only has a component which is parallel to $\vec{L}$, only the magnitude of $\vec{L}$ changes with no change in direction.

Likewise if the torque only has a perpendicular component, the direction of $\vec{L}$ changes but not the magnitude.

3 Flywheels

Consider a heavy disk rotating about the horizontal axis, loosely pivoted at O .

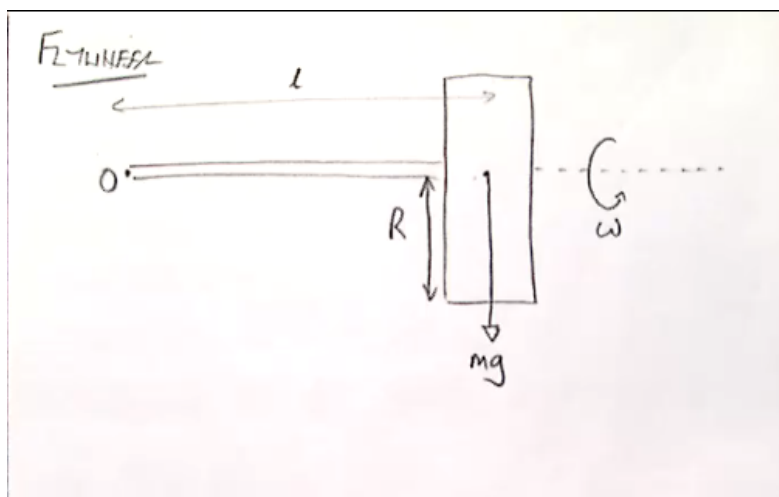


Figure 6.1

If ω is sufficiently large, the flywheel can stay horizontal instead of drooping downwards. As drawn, the $\vec{\omega}$ vector points directly outwards (leftwards).

The angular momentum vector:

$$\vec{L} = I_{\text{wheel}} \vec{\omega}$$

Where I_{wheel} is about the axis of rotation.

$$L = MR^2/2\omega$$

Which also points directly outwards. The downward force of the weight tries to drag the flywheel downwards. The torque around the origin is:

$$\vec{\tau} = \vec{r} \wedge \vec{F}$$

This creates a torque acting *into the page*, which causes the whole wheel to rotate horizontally around the origin. This results in the flywheel precessing.

Given a particular angular velocity of the wheel, ω , how quickly does this precession take place? Using $\vec{\tau} = d\vec{L}/dt$, we want to find the angular velocity of the precession, Ω .

From a top-down view, suppose the angular momentum vector at t is $\vec{L}(t)$ and after some small Δt it has new angular momentum vector $\vec{L}(t + \Delta t)$, with the vector $\Delta\vec{L}$ between them. The two angular momentum vectors make an angle $\Delta\phi$.

$$\vec{\tau} = \frac{d\vec{L}}{dt} \approx \frac{\Delta\vec{L}}{\Delta t}$$

$$|\Delta L| = |L| \sin \Delta\phi = |L| \Delta\phi \quad \text{for small angles}$$

Hence, considering magnitudes:

$$\tau = \frac{L \Delta\phi}{\Delta t} \tag{1}$$

By definition of the precession angular velocity Ω :

$$\Omega = \frac{d\phi}{dt}$$

Hence from eqn (1):

$$\Omega \approx \frac{\Delta\phi}{\Delta t} = \frac{\tau}{L}$$

We also know that:

$$L = I_{\text{wheel}} \omega = \frac{mR^2 \omega}{2}$$

And:

$$\tau = rF = lmg$$

Hence:

$$\Omega = \frac{2lmg}{\omega m R^2} = \frac{2gl}{\omega R^2}$$

3.1 Why does it not fall over?

This demonstrates precession, but doesn't explain why it actually stays up. Are there not still gravitational forces that aren't countered?

- This is not a statics problem. This is a dynamics problem with moving motion that doesn't require balanced forces, as there's a net acceleration and it's not in equilibrium.
- If the wheel tries to fall, the angular momentum vector will change in a direction where there isn't an applied torque. This fails conservation.
- Torques actually do cause motion perpendicular to the direction of the applied force.
- If the wheel isn't spinning fast enough, it will fall.

Thu 12 Feb 2026 15:00

Lecture 7 - Mechanics VII: Simultaneous Translation/Rotation & Rolling

1 Mixed Translational/Rotational Motion

For a rigid body undergoing “complex motion”, i.e. an asteroid moving through space while also rotating in an orbit, we can decompose the problem into two parts:

- Motion of the whole body, considered about the CoM (translational motion of the CoM).
- Rotations of the body about the CoM.

The result (no examinable derivation) is:

$$\vec{L} = (\vec{R} \times \vec{P})_{\text{of the CM about } O} + \left(\sum_i \vec{r}_i' \times \vec{p}_i' \right)_{\text{of components about CM}}$$

And for kinetic energy, assuming a constant angular velocity:

$$E_k = \frac{1}{2}mv_{CM}^2 + \frac{1}{2}I_0\omega^2$$

2 Sliding and Rolling

An object sliding up/down a slope is a familiar problem we’ve seen in CM1. The acceleration is given by the frictional force.

We have an object rotating with ω resulting in the centre of mass moving with v_{CM} . The object has radius r .

What is the condition for the cylindrical object to exclusively roll, with no sliding effects. Consider one point on the circumference A over one full rotation. If there has been no slipping, the distance travelled by the CoM must be exactly the circumference $2\pi r$.

The time period of one complete rotation is $T = 2\pi/\omega$. The translation speed therefore must be:

$$v = d/t = \frac{2\pi r}{2\pi/\omega} = r\omega$$

The total kinetic energy in the strictly rolling case must be:

$$E_k = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I_0\omega^2$$

Consider a coin with mass and radius m, r purely rolling along a flat surface with ω and v_{CM} .

We have already derived the moment of inertia of $I_0 = mr^2/2$. The kinetic energy is hence:

$$\begin{aligned} E_k &= \frac{1}{2}mv_{CM}^2 + \frac{1}{4}mr^2\omega^2 \\ &= \frac{1}{2}mv_{CM}^2 + \frac{1}{4}mr^2\left(\frac{v_{cm}}{r}\right)^2 \\ &= \frac{1}{2}mv_{CM}^2 + \frac{1}{4}m(v_{cm})^2 \end{aligned}$$

$$= \frac{3}{4}mv_{cm}^2$$

Suppose the coin is now rolling down an inclined slope, with incline at angle θ to the horizontal. The coin now has unbalanced forces and hence additional frictional forces f and additional translational and angular acceleration a, α

For strictly pure rolling, f provides just sufficient force to prevent slipping. This acts at the point of contact, on the coin diameter (at distance r from the CoM) and hence creates a torque about the CoM.

$$\vec{\tau} = \vec{r} \times \vec{f}$$

These are perpendicular hence:

$$\tau = rf$$

Also:

$$\tau = I_0\alpha \implies rf = I_0\alpha \implies f = \frac{I_0\alpha}{r} \quad (1)$$

We can now write the equation of motion down the slope:

$$mg \sin \theta - f = ma$$

$$mg \sin \theta - \frac{I_0\alpha}{r} = ma$$

As we're dealing with pure rolling, $v = r\omega \implies a = r\alpha \implies \alpha = a/r$:

$$mg \sin \theta - \frac{I_0a}{r^2} = ma$$

$$\implies a \left(m + \frac{I_0}{r^2} \right) = mg \sin \theta$$

$$\implies a = \frac{mg \sin \theta}{m + I_0/r^2}$$

We can simplify the problem a little by using the radius of gyration such that $I_0 = mK^2$:

$$a = \frac{mg \sin \theta}{m + mK^2/r} = \frac{g \sin \theta}{1 + K^2/r^2}$$

Assuming our object is still a uniform disk:

$$I_0 = mr^2/2 \implies K^2 = R^2/2$$

$$\implies a = \frac{g \sin \theta}{1 + 1/2} = \frac{2}{3}g \sin \theta$$

2.1 Other Objects Rolling Down Slope

Imagine a different configuration of object rolling (i.e. a ball instead of a coin). An object with the smallest I_0, K is going to be the fastest. Consider the following objects:

- Hollow Cylinder: $I_0 = mR^2$, identical to a ring.
- Solid Cylinder: $I_0 = mR^2/2$, identical to a solid coin.
- Solid Sphere: $I_0 = 2mR^2/5$.
- Hollow Sphere: $I_0 = 2mR^2/3$.

Hence the solid sphere has the smallest moment of inertia (on average has its mass distribution closest to the axis) hence will reach the bottom of the slope faster (assuming constant mass).

3 Mixed Rolling and Sliding

Imagine a very steep or slippery surface where the frictional force fails to prevent sliding and both sliding/rolling occur.

For example, in 10-pin bowling, a highly skilled bowler releases their ball perfectly in a straight line. Does the ball:

1. Roll all the way.
2. Slide all the way.
3. Slide, and then roll.
4. Roll, and then slide.

It turns out that it does (3), initially sliding before settling into a rolling motion. See next lecture for solution.

Tue 17 Feb 2026 11:00

Lecture 8 - Special Relativity I: Foundations and Time Dilation

1 Introduction

In this lecture:

- Recap from CMR1
- Einstein's two postulates
- Time dilation
- Galilean Transformations

Crucially, there are no special or absolute frames of reference - laws of motion are only sensible when we consider a frame of reference. The only slightly special frame is the one in which we are currently stationary.

Special relativity provides a theory of relative motion between inertial frames of reference. An inertial frame is one which is not accelerating relative to the other frames being considered.

2 Frames of Reference

We will generally consider two related frames of reference, denoted Σ and Σ' . Σ is our “stationary” frame, i.e. the frame taken by an observer sat on earth. Σ' is our “moving frame”, relative to the stationary observer and is moving with constant speed v .

Coordinates in the stationary frame are (x, y, z, t) , while in the moving frame they add a prime, so are given by: (x', y', z', t') .

We want to compare observations of position/experience of time/velocity in the moving frame with observations of the same categories in the stationary frame.

The frame in which an object is stationary is denoted its rest frame Σ_0 .

3 Galilean Transformations

“Galilean Invariance”: The laws of physics are invariant in all inertial (non-accelerating) frames.

“Galilean Transformation”: Time is invariant and universal in all frames.

In Classical Mechanics, we treat time as invariant, but this stops being true in special relativity. We can transform between frames in such a manner that time is kept constant, this is a Galilean Transformation, but this is not true by default.

Consider our two reference frames Σ and Σ' , where the latter moves with speed v relative to the former. Our coordinates in (x, y) become coordinates in (x', y') .

An object moves at speed u' in the x -direction in reference frame Σ' . If we assume that time is invariant (as we're still currently doing classical physics without having introduced special relativity), then:

$$t = t'$$

And as the motion is entirely in the x-axis:

$$y = y'$$

After time t :

$$x = x' + \text{distance moved by } \Sigma' \text{ in time } t \text{ relative to } \Sigma$$

$$x = x' + vt$$

The object velocity is defined as:

$$\begin{aligned} u &= \frac{dx}{dt} = \frac{d}{dt}(x' + vt) \\ &= \frac{dx'}{dt} + v = u' + v \end{aligned}$$

This holds if and only if time is the quantity we treat as being invariant. This agrees with our classical understanding.

3.1 Where does this break down?

Lets assume that the moving object is a photon, moving in Σ' with velocity c' , hence, according to the previous derivation:

$$c = c' + v$$

Therefore observers in different frames will measure different values for the speed of light...oh no!

If we assume Galilean invariance, the laws of physics are the same in all reference frames, and yet the speed of light is included as a constant in many laws (i.e. electromagnetism). Therefore observers must measure the same speed of light in all reference frames.

This is a contradiction - we cannot assume that both time and the laws of physics are invariant.

3.2 Experimental Verification

The earth travels around the sun extremely quickly, and the sun is travelling even faster around the galactic centre. Therefore, the earth is moving in space.

If Galilean relativity is correct, we would measure the speed of light in one direction as $c + u$ and measure the speed of light in the other direction as $c - u$.

This was tested by the Michelson-Morley experiment, where incoming light was split by a half-silvered mirror. Half the light travels in one direction, and half the light is split off by 90° . They reflect off a pair of mirrors and are recombined in a splitter to be observed.

If the speed of light was different in different directions, the two beams would be out of phase and we would observe an interference pattern on combination. We would expect to see phase difference that varies with time, i.e. turning from destructive to constructive etc.

This was not observed and the interference pattern generated was constant. Therefore, the results showed no variation in the speed of light.¹

4 The Solution - Einstein's Special Relativity

To fix this contradiction, Einstein came up with two postulates:

1. The laws of physics are the same in any inertial frame of reference.
 - This is the same as Galilean invariance and is easily believable.
2. The speed of light is constant in every frame of reference.
 - This was revolutionary and is much less intuitive.

The last postulate is difficult to understand intuitively, but makes everything work if we accept it as true!

¹This is the same idea used in the LIGO experiment to discover gravitational waves, except this was used to show that the path length changed, and not the speed of light. If the path lengths changed, this was due to a gravitational wave (ripple in space time) propagating to the earth and interfering with the measurement.

4.1 Proving Time Dilation

Consider an observer on the earth in frame Σ . This observer watches two rockets both travelling side by side away from the earth in the x -direction with speed v . They are both travelling in x , with a constant y -difference y_0 between them.

Suppose the upper rocket A fires a laser beam directly at B (i.e. directly in the y -direction downwards).

Frame Σ : This is the rest frame of the earth (and the observer on earth) with coordinates (x, y) .

Frame Σ' : This is the rest frame of the rockets, with coordinates (x', y') .

In the rockets rest frame Σ' , the rockets are stationary and the time taken for a laser pulse to travel between them is:

$$t_0 = \frac{y_0}{c}$$

Or equivalently:

$$y_0 = ct_0$$

In the earth's rest frame Σ , the observer on the earth doesn't just see the light travelling in the y -direction. It also sees the light moving in the x -direction, as the whole Σ' frame containing the rockets relativistically move away.

Effectively, we have:

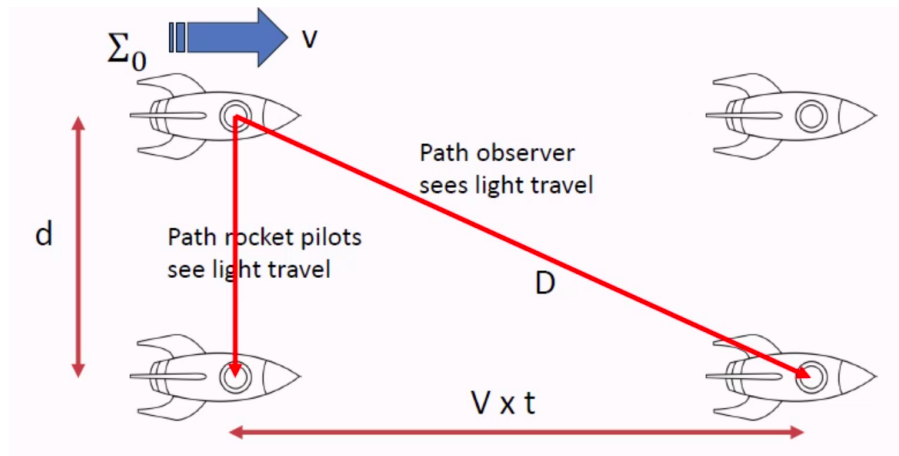


Figure 8.1

B moves distance x horizontally while it waits for the laser to hit it, and the length of the path of the laser beam in time t is given by D . If the time taken for light to reach B in Σ is t , then:

$$D = ct$$

As the speed of light is constant in all frames. As the rockets are moving away:

$$x = vt$$

This gives us a Pythagorean triangle, where:

$$D^2 = x^2 + y^2 = x^2 + y_0^2$$

And substituting in:

$$c^2 t^2 = v^2 t^2 + c^2 t_0^2$$

$$t^2 (c^2 - v^2) = t_0^2 c^2$$

$$t = t_0 \sqrt{\frac{c^2}{c^2 - v^2}}$$

Dividing through by c_2 :

$$t = t_0 \sqrt{\frac{1}{1 - \frac{v^2}{c^2}}}$$

Or:

$$t = t_0 \gamma, \quad \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$\gamma > 1$ for all object speeds that don't exceed the speed of light (which is required), so the time observed in the frame Σ is longer than the "proper time" observed by the rocket.

5 The Twin Paradox

Consider two twins, an astronaut and a physics professor. The astronaut goes on a long trip close to the speed of light and reunites with his brother. Which brother is older?

Special relativity says that because there is no absolute frame of reference, the brother travelling away also sees the same problem in reverse, i.e. he sees the physics professor brother travelling away in the opposite direction at equal and opposite speed, i.e. they are the same age.

In practice, the astronaut is non-inertial and special rel isn't sufficient, so we can't solve it directly.

Thu 19 Feb 2026 15:00

Lecture 9 - Special Relativity II: Length Contraction

1 Proving Length Contraction

Consider a horizontal laser cavity. We fire a laser from the left of the cavity (point A) to the right of the cavity (point B). The laser bounces off a mirror at point B and bounces back.

The cavity has length L_0 as measured in the rest frame Σ_0 . In this frame:

$$t_0 = \frac{2L_0}{c}, \quad \text{Where } t_0 \text{ is the time taken for the } A \rightarrow B \rightarrow A \text{ journey in } \Sigma_0$$

Now suppose the cavity moves at speed v relative to a stationary observer in a different frame (Σ).

The observer sees the cavity as having length L , how long does the $A \rightarrow B \rightarrow A$ journey take as observed by a stationary observer in Σ .

$$t = t_{A \rightarrow B} + t_{B \rightarrow A}$$

The cavity is moving away from the observer, so the laser beam doesn't just travel L when going from $A \rightarrow B$. It must travel L plus the distance the whole cavity has moved away in this time (as B is moving away from the pulse). This extra distance is $vt_{A \rightarrow B}$, hence the distance travelled when going from A to B is $L + vt_{A \rightarrow B}$.

As c is constant in all frames, we can say:

$$c = \frac{L + vt_{A \rightarrow B}}{t_{A \rightarrow B}}$$

$$t_{A \rightarrow B}(c - v) = L$$

$$t_{A \rightarrow B} = \frac{L}{c - v}$$

And now on the return journey from $B \rightarrow A$, as it travels back towards A , the cavity is moving in the same direction as the light, so A "catches up" to the pulse and results in a smaller required distance to be travelled, $L - vt_{B \rightarrow A}$

$$c = \frac{L - vt_{B \rightarrow A}}{t_{B \rightarrow A}} \implies t_{B \rightarrow A} = \frac{L}{c + v}$$

And returning to the total time:

$$\begin{aligned} t &= t_{A \rightarrow B} + t_{B \rightarrow A} \\ t &= \frac{L}{c - v} + \frac{L}{c + v} \\ &= L \left(\frac{1}{c - v} + \frac{1}{c + v} \right) \\ &= L \left(\frac{c + v + c - v}{(c - v)(c + v)} \right) \\ &= L \frac{2c}{c^2 - v^2} \\ &= \frac{2L}{c} \frac{c^2}{c^2 - v^2} \\ &= \frac{2L}{c} \frac{1}{1 - \frac{v^2}{c^2}} \end{aligned}$$

$$= \frac{2L}{c} \gamma^2$$

We now apply time dilation, which says that $t = \gamma t_0$, hence:

$$\begin{aligned} \gamma t_0 &= \frac{2L}{c} \gamma^2 \\ t_0 &= \frac{2L}{c} \gamma \end{aligned}$$

From the rest frame, we know that $t_0 = 2L_0/c$, so:

$$\frac{2L_0}{c} = \frac{2L\gamma}{c}$$

$$L_0 = L\gamma$$

As $\gamma > 1$, $\forall (u < c)$, the proper length measured from the rest frame will always be greater than the length measured by an observer, hence length is contracted for a moving object.

2 Example

In a particle accelerator, protons are accelerated to a measly $0.9c$. These protons pass through a tunnel of length 2km as viewed from the a laboratory rest frame at the accelerator.

Recall that:

$$\begin{aligned} t &= \gamma t_0 \\ L &= \frac{L_0}{\gamma} \\ \gamma &= \frac{1}{\sqrt{1 - \beta^2}}, \quad \text{where: } \beta = u/c \end{aligned}$$

2.1 How long would the journey take, according to the rest frame?

Viewed in the lab frame, and $v = d/t$ so:

$$t = \frac{2 \times 10^3 \text{m}}{0.9 \times 3 \times 10^8} = 7.4 \times 10^{-6} \text{s} = 7.4 \mu\text{s}$$

No relativity needed!

2.2 How long would the journey take, according to the proton's frame?

We do need to consider relativity here, and use time dilation:

$$\begin{aligned} t &= \gamma t_0 \\ \gamma &= \frac{1}{\sqrt{1 - 0.9^2}} = 2.3 \\ t = \gamma t_0 &\implies t_0 = \frac{t}{\gamma} = \frac{7.4 \mu\text{s}}{2.3} = 3.2 \mu\text{s} \end{aligned}$$

2.3 How long is the tunnel, according to the proton's frame?

The rest frame now corresponds to the tunnel. It has proper length in its rest frame of 2km.

As viewed by the protons in their rest frame, it is the tunnel that is moving and is rushing towards them at $0.9c$. This is therefore a length contraction problem.

$$L = \frac{L_0}{\gamma} = \frac{2 \text{km}}{2.3} = 870 \text{m}$$

2.4 Checking Values

To check, we can reuse $v = d/t$ but this time in the proton rest frame.

$$d = 870\text{m}, \quad t = 3.2 \times 10^{-6}\text{s}$$

Hence

$$v = \frac{870\text{m}}{3.2 \times 10^{-6}\text{s}} = 2.7 \times 10^8 \text{ms}^{-1} = 0.9c \text{ as required!}$$

3 Example II: Cosmic Rays

The highest energy cosmic rays are protons with massive energies $E \sim 10^{20}\text{eV}$, compared to the LHC with $E \sim 10^{12}\text{eV}$

This corresponds to a $\gamma = 10^{11}$, and our galaxy is $\sim 10^{20}\text{m}$ across.¹

As viewed on earth, these cosmic protons travel at $v \approx c$, hence,

$$t \approx \frac{d}{v} \approx \frac{10^{20}}{3 \times 10^8} = 3 \times 10^{11}\text{s} \approx 10^4 \text{years}$$

As viewed by the protons:

$$t_0 = \frac{t}{\gamma} = \frac{3 \times 10^{11}}{10^{11}} \approx 3\text{s}$$

Which is starkly different!

4 Lorentz Transformations

In general, transforming between coordinates in two different frames can get extremely messy, more than can easily be handled in simple applications of length contraction or time dilation.

The Lorentz Transformations provide a general set of coordinate transformations between Σ and Σ' , i.e. $(x, y, z, t) \rightarrow (x', y', z', t')$.

These transformations must be:

1. Symmetric about a change in sign of u , i.e. the transformation from $\Sigma \rightarrow \Sigma'$ with u and the transformation from $\Sigma' \rightarrow \Sigma$ with $-u$ must be the same.
2. They must be linear as otherwise a constant speed in one would not be constant in the other (and therefore not inertial frames).
3. If relative velocity is only in the x direction, the transformations must be independent of y and z .

These transformations are (assuming motion only in x):

$$\begin{aligned} x &= \gamma(x' + vt') \\ y &= y' \\ z &= z' \\ t &= \gamma\left(t' + \frac{x'v}{c^2}\right) \end{aligned}$$

And:

$$\begin{aligned} x' &= \gamma(x - vt) \\ y' &= y \\ z' &= z \\ t' &= \gamma\left(t - \frac{xv}{c^2}\right) \end{aligned}$$

We do not need to know the derivations, but do need to know the results.

¹ $\gamma = E/mc^2$ - will be expanded on in later lectures.

Tue 24 Feb 2026 11:00

Lecture 10 - Special Relativity III: Lorentz Transformations

Recap of the first two relativity lectures:

- Recap from Semester 1 Special Rel:
 - Background
 - Einstein's postulates.
 - Time dilation.
 - Lorentz contraction.
- Galilean transformations and when they fail.
- Lorentz transformations for space and time.

1 Galilean vs. Lorentz Transformations

In a standard setup with two frames, Σ and Σ' , where Σ' moves with speed v relative to Σ .

In Σ , we have coordinates (x, y, t) , in Σ' we have coordinates (x', y', t') .

In a Galilean transformations, we kept time invariant across both frames, so $t = t'$. This lead to $x' = x - vt \implies u' = u - v$ but raised a contradiction with the speed of light as observers in different frames would take two measurements of the speed of light.

In the second lecture, we consider the speed of light as being invariant across all frames instead of time to resolve the contradiction. This gave us the Lorentz Transformations:

$$x' = \gamma(x - vt)$$

$$t' = \gamma\left(t - \frac{xv}{c^2}\right)$$

Where $y = y', z = z'$.

As we cannot use $u' = u - v$ due to the contradiction with the speed of light, we in this lecture will consider:

- Lorentz transformations for velocity.
- Space-time, Lorentz invariance, causality.

2 Lorentz Transformation for Velocity

We can revisit the question from the first lecture, but in the Lorentz transformation perspective rather than a Galilean perspective.

Again, we have our two standard frames Σ, Σ' . An object is moving with speed u' in frame Σ' . We want to find an expression for the velocity of the object as viewed from Σ , given by u . The prime frame moves at speed v wrt the rest frame and the objects motion is entirely along the x axis.

By definition, we have:

$$u = \frac{dx}{dt} = \frac{dx}{dt'} \frac{dt'}{dt} = \frac{\left(\frac{dx}{dt'}\right)}{\left(\frac{dt}{dt'}\right)} \quad (1)$$

And:

$$\frac{dx}{dt'} = \gamma \left(\frac{dx'}{dt'} + v \right) = \gamma (u' + v) \quad (2)$$

$$\frac{dt}{dt'} = \gamma \left(1 + \frac{v}{c^2} \frac{dx'}{dt'} \right) = \gamma \left(1 + \frac{u'v}{c^2} \right) \quad (3)$$

And substituting (3) and (2) into (1):

$$u = \frac{\gamma(u' + v)}{\gamma \left(1 + \frac{u'v}{c^2} \right)} \Rightarrow \boxed{u = \frac{u' + v}{1 + \frac{u'v}{c^2}}}$$

To get the reverse transformation, we invoke forward-backward symmetry, i.e. swap $x \rightarrow x'$, $t \rightarrow t'$ etc, where $v \rightarrow -v'$:

$$u' = \frac{u - v}{1 - \frac{uv}{c^2}}$$

2.1 Limiting Cases

Firstly, we'll look at the non-relativistic case, i.e. where $u, v \ll c$:

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}} = \frac{u' + v}{1 + \text{small}} \approx u' + v$$

This is the classical Galilean approach, which is what we'd expect for non-relativistic behaviour.

What about an ultra-relativistic limit where $u, u' \rightarrow c$ with $v \ll c$. Now we have:

$$u = \frac{u' + \text{small}}{1 + \text{small}} \approx u'$$

As the speed of light is invariant, speeds close to the speed of light do not change between reference frames.

2.2 Example

An observer on a rocket sees a second rocket moving away from them at $u_1 = 0.8c$. Another observer on a planet sees the first rocket moving at $0.7c$. Assuming all motion is in the same direction, how fast does the planetary observer see the second rocket?

- From Σ' (the first rocket's rest frame), the second rocket moves with $0.8c$.
- From Σ (the planetary rest frame), the first rocket moves with $0.7c$, and the second rocket moves with some unknown speed u .

We apply the formula:

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$$

In Σ' , $u' = 0.8c$. The two frames move with relative velocity equal to the change in speed of the first rocket from $\Sigma \rightarrow \Sigma'$: $v = 0.7c$.

In frame Σ we are trying to do:

$$\begin{aligned} u &= \frac{0.8c + 0.7c}{1 + \frac{0.8c \times 0.7c}{c^2}} \\ &= \frac{1.5c}{1.56} = 0.96c \end{aligned}$$

3 Space-Time and Lorentz Invariants

From length contraction and time dilation, we can see that in special relativity, intervals in time and space are not fixed but vary from frame-to-frame.

This means that the order of events may even be different for different observers.

There are some quantities however which are invariant across frames, such as the speed of light. One of these is space-time.

3.1 Space-Time Invariance

As usual, we have Σ and Σ' with relative speed v . The axes of these frames coincide at $t = 0, x = 0$ and $t' = 0, x' = 0$.

At $t = t' = 0$, there is a flash of light at the origin which propagates isotropically.

In Σ , the light forms a spherical shell of radius Δr at time Δt , and in Σ' it reaches a radial distance $\Delta r'$ at time $\Delta t'$.

After time ΔT in Σ we have $\Delta r = c\Delta T$. Similarly in Σ' we have $\Delta r' = c\Delta t'$.

Squaring these we have:

$$\begin{aligned}(\Delta r)^2 &= c^2 \Delta t^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 \\ (\Delta r')^2 &= c^2 \Delta t'^2 = \Delta x'^2 + \Delta y'^2 + \Delta z'^2\end{aligned}$$

Hence:

$$c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 = 0 = c^2 \Delta t'^2 - \Delta x'^2 - \Delta y'^2 - \Delta z'^2$$

This means that, **for light**:

$$c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 = 0$$

We call this quantity Δs^2 :

What about for objects moving at a speed less than the speed of light? We want to apply the Lorentz transformations to $\Delta s'^2$ (again treating the “boost” as being in x):

$$\begin{aligned}\Delta s'^2 &= c'^2 \Delta t'^2 - \Delta x'^2 - \Delta y'^2 - \Delta z'^2 \\ &= c^2 \gamma^2 \left(\Delta t - \frac{v}{c^2} \Delta x \right)^2 - \gamma^2 (\Delta x - v \Delta t)^2 - \Delta y^2 - \Delta z^2 \\ &= c^2 \gamma^2 \left(\Delta t^2 + \frac{v^2}{c^4} \Delta x^2 - 2 \frac{v}{c^2} \Delta t \Delta x \right) - \gamma^2 (\Delta x^2 + v^2 \Delta t^2 - 2v \Delta x \Delta t) - \Delta y^2 - \Delta z^2 \\ &= \Delta t^2 (c^2 \gamma^2 - v^2 \gamma^2) + \Delta x^2 \left(\frac{v^2 \gamma^2}{c^2} - \gamma^2 \right) + \Delta t \Delta x (-2v \gamma^2 + 2v \gamma^2) - \Delta y^2 - \Delta z^2 \\ &= \Delta t^2 \gamma^2 (c^2 - v^2) + \Delta x^2 \gamma^2 \left(\frac{v^2}{c^2} - 1 \right) - \Delta y^2 - \Delta z^2 \\ &= \Delta t^2 \frac{c^2 - v^2}{1 - \frac{v^2}{c^2}} + \Delta x^2 \frac{\left(\frac{v^2}{c^2} \right) - 1}{1 - \frac{v^2}{c^2}} - \Delta y^2 - \Delta z^2 \\ \Delta s'^2 &= \boxed{c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 = \Delta s^2}\end{aligned}$$

Hence Δs^2 is invariant under the Lorentz transformations regardless of speed.

Thu 26 Feb 2026 15:00

Lecture 11 - Mechanics VIII: 10-Pin Bowling, Impulse and Centre of Percussion

1 10-Pin Bowling

Consider a ball of mass M and radius R being bowled in an alley. The ball is released with velocity v_0 . As it makes contact with the wooden floor (coefficient of friction μ) the ball slows down and begins to rotate.

Initially, it is both sliding and rolling. After some time, the ball ceases sliding and rolls only.

Write expressions for:

1. Linear deceleration.
2. Angular acceleration.
3. The speed of the centre of the ball after time t .
4. The angular velocity of the ball about its centre after time t .

Then:

5. Show that the ball with both slide and rotate until time t_{roll} elapses, where:

$$t_{\text{roll}} = \frac{v_0}{\mu g \left(1 + \frac{R^2}{K^2}\right)}$$

6. After time t_{roll} the ball rolls at constant speed v_{roll} . Give an expression for v_{roll} .

1.1 Part I

The ball experiences three forces:

- Weight force mg .
- Normal reaction force N .
- Frictional force f .

The first two forces cancel, hence the net force is simply friction:

$$f = -\mu R = -\mu mg = ma \implies a = -\mu g$$

1.2 Part II

The friction causes the ball to pick up an angular velocity as it rotates, by supplying a torque:

$$\tau = I_0 \alpha = |\vec{r} \times \vec{f}|$$

Hence:

$$I_0 \alpha = r|f|$$

$$\alpha = \frac{\mu mgr}{I_0}$$

In terms of the radius of gyration, $I_0 = mK^2$:

$$\alpha = \frac{\mu mgr}{mK^2} = \frac{\mu gr}{K^2}$$

1.3 Part III

The linear deceleration is constant, so we can just use SUVAT:

$$v = u + at$$

$$V_{CM}(t) = v_0 - \mu g t$$

1.4 Part IV

The angular acceleration α is also a constant. We can use the angular equivalent of SUVAT:

$$\omega = \omega_0 + \alpha t$$

$$\omega(t) = \omega_0 + \frac{\mu g r t}{K^2}$$

And as it starts purely sliding $\omega_0 = 0$

$$\omega(t) = \frac{\mu g r t}{K^2}$$

1.5 Part V

Pure rolling defines the moment when the sliding stops and the motion is exclusively rotational. This has the condition that the frictional force totally counteracts sliding motion and hence $v_{CM}(t) = r\omega(t)$. Equating these:

$$v_0 - \mu g t = \frac{\mu g r^2 t}{K^2}$$

$$t_{\text{roll}} \left(\frac{\mu g r^2}{K^2} + \mu g \right) = v_0$$

$$t_{\text{roll}} = \frac{v_0}{\mu g (1 + r^2/K^2)}$$

as required!

For a spherical object, this simplifies to:

$$\frac{7v_0}{7\mu g}$$

1.6 Part VI

When $t = t_{\text{roll}}$, we can use the time dependence of v :

$$v_{\text{roll}} = v_{CM}(t_{\text{roll}}) = v_0 - \mu g \left(\frac{2v_0}{7g\mu} \right)$$

$$= v_0 \left(1 - \frac{2}{7} \right)$$

$$v_{\text{roll}} = \frac{5}{9} v_0$$

2 Impulsive Forces and Torques (“Hitting Things”)

An impulse is a force that acts for a very short period. This causes a change in the translational velocity/momentum according to:

$$\frac{d\vec{p}}{dt} = \vec{F}$$

$$\Delta\vec{p} = \int \vec{F} dt = \vec{F} \Delta T$$

For a body which can rotate, there's also an impulsive torque applied:

$$\frac{d\vec{L}}{dt} = \tau$$

$$\Delta\vec{L} = \int \vec{\tau} dt = |\vec{r} \times \vec{F}| \Delta t$$

Broadly, we can break the impacts of an impulsive force on a rigid body to:

- A change in CM translational velocity as Δv_{CM} given from $\Delta p = m\Delta v$.
- A change in angular velocity about the CM fby $\Delta\omega$ given by $\Delta L = I\Delta\omega$

There are some special ways to hit things with special cases of these changes, however.

Force at CoM

Consider a rigid body of mass M and an impulsive force $\vec{F}$ is applied at the centre of mass for short time Δt .

$$F = \frac{dp}{dt} \therefore \Delta p = F\Delta t$$

$$m\Delta v = F\Delta t$$

So the whole body picks up a new translational velocity:

$$\Delta v = \frac{F\Delta t}{m} \quad (1)$$

In this case, there is no torque about the centre of mass ($r = 0$), so no induced rotation. The whole body sets off at the new velocity without rotating.

Force Below CoM

Now consider a force being applied below the object's CoM. Suppose it's applied at distance b directly below the CoM.

Just as done before, the whole body has an induced translational motion. This translational motion happens at the same Δv .

There is now however some torque applied about the CoM.

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\tau = bF$$

We also know that:

$$\tau = \frac{d\vec{L}}{dt} \implies \boxed{\Delta L = \tau \Delta t = bF\Delta t}$$

Finally, $L = I_0\omega$:

$$bF\Delta t = I_0\Delta\omega \implies \boxed{\Delta\omega = \frac{bF\Delta t}{I_0}} \quad (2)$$

What is the combined impact of this combined translational and rotational induced motion on some point *above* the CoM? Suppose this point A , lies in line with the CoM vertically, at height h .

This point receives induced velocity Δv and some induced $\Delta\omega$ rotationally. For this point above, the rotation is in the opposite direction, causing the upper section to twist towards the induced force (and the lower section to be pushed away by the force).

Hence at A , Δv and $\Delta\omega$ act in opposite directions. For some specific choices of a, h , we can cause these to cancel at A .

By $v = r\omega$, this gives the conditions:

$$\Delta v = h\Delta\omega$$

$$\frac{F\Delta t}{m} = h \frac{rbF\Delta t}{I_0}$$

$$\frac{1}{m} = h \frac{hb}{I_0}$$

This leaves a problem which is purely about the geometry of the shape and point and not the force:

$$b = \frac{I_0}{mh}$$

What if point A is no longer hypothetically free to move and is instead a pivot. In a bat or a racquet, this is the point at which the object is being held by the person.

If this is true, and a force is applied at the lower point B , the hand feels no reaction force if:

$$b = \frac{I_0}{mh}$$

This is called the centre of percussion or the sweet spot.

2.1 Uniform Rod, Pivot at One End

Consider our object is now a bat, modelled as a slender rod with a pivot at one end. The CoM sits at distance h from the pivot and the CoP sits at $h + b$. The total length is $L = 2h$, assuming the bar is uniform.

The total mass is M . As previously derived, the moment of inertia through its centre of mass is:

$$I_0 = \frac{ML^2}{12}$$

Hence:

$$\begin{aligned} b &= \frac{ML^2}{12Mh} \\ &= \frac{2ML^2}{12ML} \\ &= \frac{L}{6} \end{aligned}$$

Hence the CoP sits at $L/2 + L/6 = \frac{2L}{3}$ below the pivot.

2.2 Door Stop

We can place the a door stop at a length $2L/3$ away from the hinges to minimise (theoretically eliminate) the force felt by the pivot (the hinges). This minimises the force felt by the hinges and gives the smoothest possible effect.

3 Rigid Penduli

he centre of percussion equivalently defines the Centre of Oscillation. This means that if we have a complex object acting as a pendulum, the period of oscillation will be equal to the period of oscillation of a simple pendulum with a point mass at the CoP.

Tue 03 Mar 2026 11:00

Lecture 12 - Special Relativity IV: More Space-Time

Recap from last lecture:

- Lorentz Transformations for space and time.
- Lorentz Transformations for velocity/velocity addition.
- A Lorentz Invariant: The space-time interval is invariant across frames of reference:

$$\delta S^2 = c^2 \Delta t^2 - \Delta r^2$$

In this lecture:

- Minkowski space and causality.
- Relativistic energy and momentum.

1 Space-Time Invariance

Last lecture we showed that:

$$c^2 \Delta t^2 - \Delta r^2 = c^2 \Delta t'^2 - \Delta r'^2 = \quad (= 0 \text{ for light})$$

And we define the space-time interval Δs as $\Delta S^2 = c^2 \Delta t^2 - r^2$. There is some relationship between space and time that keeps this quantity constant across frames. What does this quantity actually *mean*, however?

In a 3D set of axes, we have:

$$\Delta r^2 = \Delta x^2 + \Delta y^2 + \Delta z^2$$

And we have defined:

$$\Delta s^2 = c^2 \Delta t^2 - \Delta r^2 = c^2 \Delta t^2 - (\Delta x^2 + \Delta y^2 + \Delta z^2)$$

This looks almost like a 4D space (where one axis is time), but with some strange negative signs appearing? We don't have the equipment yet to properly understand the negative signs, but can be in future courses.

2 Minkowski Space (Space-Time Diagrams)

In 2D, we consider one spatial coordinate and one time coordinate, where the x axis represents our spatial coordinate and the vertical axis is given by ct .

Consider some kind of light-emitting event happening at the origin. This light travels at a fixed speed, so we end up with the path of light at 45 degrees on this space-time diagram:

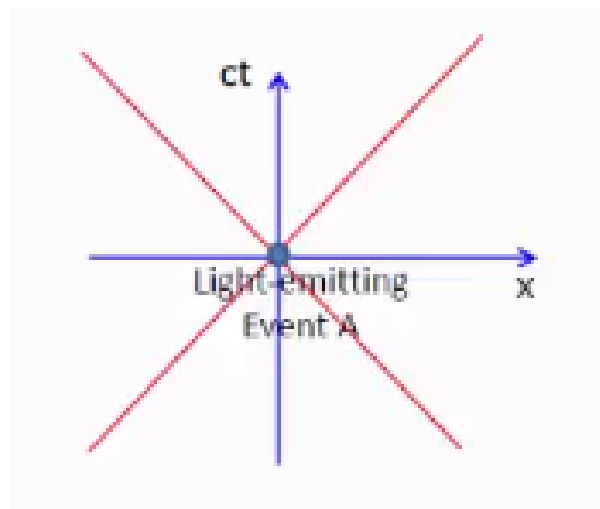


Figure 12.1

These are given by the lines $c^2t^2 = \pm x^2$ and are called “light cones”, as they form cones in higher dimensions. The positive velocity axis is called the “future cone”, the path taken by light emerging from event A. The negative velocity axis is called the “past cone”, and represents the path taken by light emerging at A.

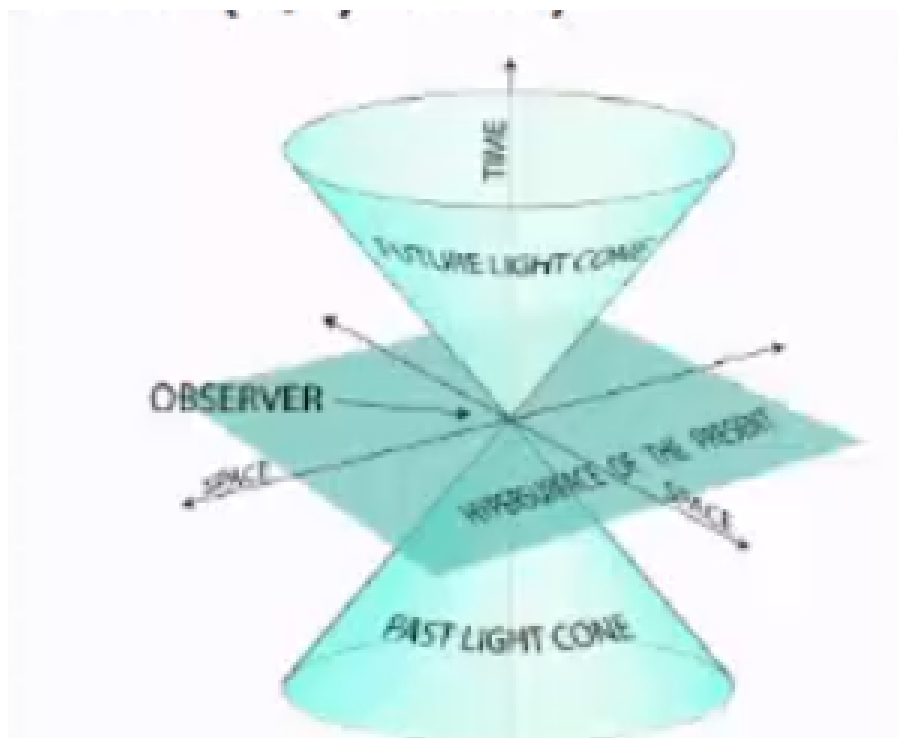


Figure 12.2: Light cones in a 3D space (two spatial, one velocity)

Inside the future cone, events may be caused by the event happening at A. Things travelling at a velocity $< c$ can reach them.

The region inside the past cone gives regions in spacetime where events could have caused A.

Events outside the two cones cannot influence, nor can they be influenced by A. They cannot be causally connected in any way to the event at A.

We now have two events. A is defined as before, and a new event B separated from A by distance $\Delta \underline{r}$ and time Δt .

What is the condition that B lies within the light cone given by A? We require:

$$|\Delta \underline{r}| \leq c|\Delta t|$$

Or in invariant form:

$$\Delta S^2 = c^2 \Delta t^2 - \Delta r^2 \geq 0$$

We call ΔS the space-time interval, or invariant distance.

There are three possible cases:

- $\Delta S^2 > 0$, “time-like intervals”. This is where $|\Delta \underline{r}| \leq c|\Delta t|$ and B is inside A’s light cone. $\Delta S \in \mathbb{R}$. Here, a signal can pass between A and B and the two events may be causally connected (where A causes B).
- $\Delta S^2 = 0$. Here, $|\Delta \underline{r}| = c|\Delta t|$. This is where B sits on A’s light cone, and A/B may be causally connected. A signal can only travel between the two events if the signal is at velocity c . This is called a “light-like interval”
- $\Delta S^2 < 0$, “space-like intervals”. This is where $|\Delta \underline{r}| > c|\Delta t|$. $\Delta S = \sqrt{c^2 \Delta t^2 - \Delta r^2} \in i\mathbb{R}$. No signal can pass between A and B, not even light. No causal connection is possible.

3 Causality and Time Order

It is possible for observers in two different frames to disagree on the order in which events have taken place. If we have two events with t_1, t_2 in Σ and t'_1, t'_2 in Σ' , there are possible cases where:

- $t_2 > t_1$.
- $t'_1 > t'_2$.

This may seem uncomfortable - does this cause a breakdown in causality?

Suppose we have a frame Σ with two events, $(x_1, t_1), (x_2, t_2)$, such that $\Delta t = t_2 - t_1 > 0$.

Frame Σ' moves with speed v relative to Σ and in Σ' the same two events have coordinates (x'_1, t'_1) and (x'_2, t'_2) . For the observers to disagree on causality, we have $\Delta t' t'_2 - t'_1 < 0$. We begin by applying the Lorentz Transformation to $\Delta t'$:

$$\begin{aligned} \Delta t' &= \gamma \left(t_2 - \frac{v}{c^2} x_2 \right) - \gamma \left(t_1 - \frac{v}{c^2} x_1 \right) < 0 \\ \implies \gamma(t_2 - t_1) + \frac{\gamma v}{c^2} (-x_2 + x_1) &< 0 \\ \Delta t - \frac{v}{c^2} \Delta x &< 0 \end{aligned}$$

I.e. $\Delta t < \frac{v}{c^2} \Delta x$ or $c\Delta t < \frac{v}{c} \Delta x$. We know that $v/c < 1$, hence $c\Delta t < \Delta x$. This means that:

$$\Delta S = \sqrt{c^2 \Delta t^2 - \Delta x^2}$$

Which is the square root of a negative number, hence this gives a space-like interval. This means that while we can define two points in space-time where two observers will disagree about their order, the space-time invariant is imaginary hence these events cannot be causally linked. This means that observers disagreeing on event order does not break causality.

4 Simultaneity

Consider a relativistic ladder passing through a stationary garage. The ladder’s proper length is longer than the proper length of the garage, so the ladder could not fit inside the garage if both were stationary.

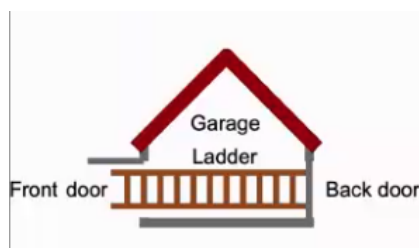


Figure 12.3

However, viewed from the rest frame of the garage, the ladder is length contracted. From this frame, the ladder can momentarily (before smashing into the back door) fit entirely in the garage and it would even be possible to briefly close the front door.



Figure 12.4

From the ladder's rest frame however, the garage is length contracted and not the ladder. Therefore, from this frame, there's no possible way to fit the ladder into the garage.

How can it simultaneously be possible and impossible depending on the rest frame being considered?

4.1 Problem

Let the ladder have proper length $l_0^L = 20\text{m}$ and the garage have proper length $l_0^G = 10\text{m}$. The ladder moves at $0.9c$ wrt the garage.

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} = \frac{1}{\sqrt{1 - 0.9^2}} \approx 2.29$$

In the garage's rest frame, the ladder length is $L^L = \frac{20}{2.29} = 8.7\text{m}$. So, from the garage's POV, the ladder fits!

We now consider the length of the garage from the ladder's rest frame: $L^G = \frac{10}{2.29} = 4.4\text{m}$, so from the ladder's POV, the ladder cannot fit! This creates a paradox.

4.2 Solution

We can resolve this problem because the concept of simultaneity is frame dependant. There are two things that happened here:

- A. The front of the ladder reached the back garage door (the first figure)
- B. The back of the ladder reaches the front garage door.

We've been assuming that these events occur at exactly the same time, simultaneously, but this isn't the case.

The time taken between these events has to obey a Lorentz transformation. If they are the same in some frame (i.e. the garage, for the ladder to fit perfectly inside), then $\Delta t_{AB} = 0$. In the rest frame of the ladder:

$$\Delta t'_{AB} = \gamma \left(\Delta t_{AB} - \frac{v}{c^2} \Delta x_{AB} \right) = \frac{\gamma v}{c^2} \Delta x_{AB} \neq 0$$

Hence, if the garage perceives the events as being simultaneous, they cannot be observed as being simultaneous in the ladder's frame.

Thu 5 Mar 2026 15:00

Lecture 13 - Orbits I: Formalism of Orbits

1 Introduction to Orbits

What is an Orbit?

Orbit $\equiv$ motion of an object in a plane (2D only) under the influence of a force. Generally, this is a closed repeating path (especially for this course) but this is not always true.

For orbits, we limit ourselves to forces and potentials which are “Central Conservative”.

- Central Forces:

- A central force has the form $\underline{F} = f(r)\hat{r}$. These forces act only in a radial direction and vary only as a function of radial distance. These forces conserve angular momentum.

- For a central force:

$$\underline{F} = f(r)\hat{r} \implies \underline{\tau} = \underline{r} \times \underline{F} = f(r)\underline{r} \times \hat{r} = 0$$

Hence there is no torque. Since torque is the derivative of angular momentum, angular momentum is conserved.

- Also, since $\underline{L} = \underline{r} \times \underline{p} = \underline{r} \times m\dot{\underline{r}} = r \times m\dot{\underline{r}}$, $\underline{r} \times \dot{\underline{r}}$ is conserved. As $\underline{r}$ and $\dot{\underline{r}}$ are both perpendicular to $\underline{L}$, they sit in the same 2D plane.

This confirms that the orbits are exclusively 2D planes.

- Conservative Forces:

- A conservative force is one that conserves total energy ($E_k + E_{\text{potential}}$). The net work done by returning to the start point via a closed path (most orbits) is zero.

$$\oint \underline{F} \cdot d\underline{r} = 0$$

- For every conservative force, there exists a scalar potential which satisfies:

$$\underline{F} = -\nabla V = -\frac{\partial V}{\partial r}$$

We will therefore choose to work with forces and potentials interchangeably. Dealing with potentials is often easier due to them being a scalar.

This includes all non-dissipative forces, i.e. gravity, electrostatics, the strong nuclear force etc. It excludes dissipative forces where energy is lost, such as friction.

1.1 Polar Coordinates

As we've established that coordinates exist purely in 2D, it makes the most sense to express them in polar coordinates, with the origin at the orbit centre.

Instead of being interested in unit vectors $\hat{x}, \hat{y}$ we're interested in unit vectors $\hat{r}$ pointing radially outwards and $\hat{\theta}$ pointing tangentially in the positive direction of motion (anticlockwise).

Given an arbitrary point:

$$(x, y) = (r, \theta)$$

We can define:

$$\underline{r} = r \underline{\hat{r}} \quad (1)$$

And:

$$\begin{aligned} \underline{\hat{r}} &= \cos \theta \underline{\hat{x}} + \sin \theta \underline{\hat{y}} \\ \underline{\hat{\theta}} &= -\sin \theta \underline{\hat{x}} + \cos \theta \underline{\hat{y}} \end{aligned}$$

We want to be able to take their time derivatives:

$$\begin{aligned} \frac{d\underline{\hat{r}}}{dt} &= \frac{d\theta}{dt} \frac{d}{d\theta} (\cos \theta \underline{\hat{x}} + \sin \theta \underline{\hat{y}}) \\ &= \dot{\theta} (-\sin \theta \underline{\hat{x}} + \cos \theta \underline{\hat{y}}) \\ &\Rightarrow \frac{d\underline{\hat{r}}}{dt} = \dot{\theta} \underline{\hat{\theta}} \end{aligned} \quad (2)$$

By the same logic:

$$\begin{aligned} \frac{d\underline{\hat{\theta}}}{dt} &= \frac{d\theta}{dt} \frac{d}{d\theta} (-\sin \theta \underline{\hat{x}} + \cos \theta \underline{\hat{y}}) \\ &\Rightarrow \frac{d\underline{\hat{\theta}}}{dt} = -\dot{\theta} \underline{\hat{r}} \end{aligned} \quad (3)$$

These let us handle cases where the distances from the axes are changing (i.e. object in an elliptical orbit). The derivation is non-examinable and the results are in the formula sheet.

2 Motion of an Object in Central Conservative Potential

Assume an elliptical orbit. The origin of the coordinate system is at the left focus on the ellipse (the sun) and a planet is in orbit.

The planet sits at varying distance r from the sun and has angle from the positive horizontal θ .

2.1 Planetary Velocity

By definition:

$$\begin{aligned} \underline{v} &= \dot{\underline{r}} = \frac{d}{dt} \underline{r} = \frac{d}{dt} (r \underline{\hat{r}}) \\ \underline{v} &= r \frac{d\underline{\hat{r}}}{dt} + \frac{dr}{dt} \underline{\hat{r}} = r \dot{\theta} \underline{\hat{\theta}} + \dot{r} \underline{\hat{r}} \end{aligned}$$

These two components give the tangential and then the radial velocity.

2.2 Planetary Acceleration

To find the acceleration, we take the time derivative again. Because r , $\dot{r}$, $\dot{\theta}$, and the unit vectors themselves are all functions of time, we use the chain rule with respect to t .

$$\begin{aligned} \underline{a} &= \frac{d}{dt} (\dot{r} \underline{\hat{r}}) + \frac{d}{dt} (r \dot{\theta} \underline{\hat{\theta}}) \\ \underline{a} &= (\ddot{r} \underline{\hat{r}} + \dot{r} \dot{\underline{\hat{r}}}) + (\dot{r} \dot{\theta} \underline{\hat{\theta}} + r \ddot{\theta} \underline{\hat{\theta}} + r \dot{\theta} \dot{\underline{\hat{\theta}}}) \end{aligned}$$

Substituting the unit vector time derivatives, $\dot{\underline{\hat{r}}} = \dot{\theta} \underline{\hat{\theta}}$ and $\dot{\underline{\hat{\theta}}} = -\dot{\theta} \underline{\hat{r}}$:

$$\underline{a} = \ddot{r} \underline{\hat{r}} + \dot{r} (\dot{\theta} \underline{\hat{\theta}}) + \dot{r} \dot{\theta} \underline{\hat{\theta}} + r \ddot{\theta} \underline{\hat{\theta}} + r \dot{\theta} (-\dot{\theta} \underline{\hat{r}})$$

And factorising by unit vectors:

$$\underline{a} = (\ddot{r} - r \dot{\theta}^2) \underline{\hat{r}} + (r \ddot{\theta} + 2\dot{r} \dot{\theta}) \underline{\hat{\theta}}$$

This gives us a radial component of acceleration and a tangential component of acceleration.

3 Forces

We know that $\underline{F} = m\underline{a}$ must apply, and we can resolve this out into radial and tangential components.

In the radial direction:

$$f(r) = m(\ddot{r} - r\dot{\theta}^2) \quad (4)$$

As the force is central conservative, there can be no component in the $\underline{\hat{\theta}}$ direction. Hence:

$$r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 \quad (5)$$

4 Angular Momentum

We again know that $\underline{L} = \underline{r} \times \underline{F} = m\underline{r} \times \underline{v}$. Let the modulus of the angular momentum vector being $l = |\underline{L}|$.

Only the tangential component of the velocity has a cross product with the radius, hence we have:

$$l = mrv_t$$

Where v_t is the tangential portion of the velocity. Inserting the tangential velocity:

$$l = m\underline{r} \times r\dot{\underline{\hat{\theta}}} = mr^2\dot{\underline{\hat{\theta}}}$$

As an aside, noting that $dl/dt = mr^2\ddot{\underline{\hat{\theta}}} + 2mr\dot{r}\dot{\underline{\hat{\theta}}}$:

$$\frac{dl}{dt} = mr(r\ddot{\underline{\hat{\theta}}} + 2\dot{r}\dot{\underline{\hat{\theta}}})$$

This is equal to mrF_T . We have already shown that $F_T = 0$, hence l is not time dependant, so is conserved.

5 Total Energy

As the field is conservative, it must be a constant equal to the sum of the kinetic and potential components.

$$E = E_k + E_p = \frac{1}{2}m\underline{v}^2 + V(r)$$

Where $V(r)$ is the central potential where $\underline{F} = -\frac{\partial V}{\partial \underline{r}}\underline{\hat{r}}$. Using the known value for velocity:

$$E = \frac{1}{2}m(r\dot{\underline{\hat{\theta}}} + \dot{r}\underline{\hat{r}})^2 + V(r)$$

Noting that a unit vector dotted with themselves is one, and dotted with the (perpendicular) other gives zero:

$$E = \frac{1}{2}m(r^2\dot{\underline{\hat{\theta}}}^2 + \dot{r}^2) + V(r)$$

Where all the cross terms become zero. We can eliminate $\dot{\underline{\hat{\theta}}}$ by using:

$$l = mr^2\dot{\underline{\hat{\theta}}} \implies \dot{\underline{\hat{\theta}}} = \frac{l}{mr^2}$$

Hence:

$$E = \frac{1}{2}m\left(r^2\frac{l^2}{m^2r^4} + \dot{r}^2\right) + V(r)$$

$$\boxed{E = \frac{1}{2}\frac{l^2}{mr^2} + \frac{1}{2}m\dot{r}^2 + V(r)}$$

Note that:

- The first term is the kinetic energy of the tangential motion. In the case of a circle, this gives $0.5l^2/I_0 = 0.5I^2\omega^2/I = \frac{1}{2}I\omega^2$ which is as expected.
- The second term is zero for circular motion where the radius is constant.

Tue 10 Mar 2026 11:00

Lecture 14 - Special Relativity V: Relativistic Mechanics

Recap of last lecture:

- Minkowski Space (space-time)
- Time-like and space-like intervals.
- Ensuring that this new model does not violate causality.
- Simultaneity.

1 Relativistic Momentum

We generally use:

$$\mathbf{p} = m\mathbf{v}$$

But this is just an approximation for $v \ll c$. This would:

- Allow $v > c$ for an object with sufficient momentum.
- Not allow massless objects like photons to have any momentum, but we know this is true.
- Is not obviously conserved.

Instead, we (proof non-examinable) use:

$$\mathbf{p} = \gamma m\mathbf{v}$$

This is the “true” expression for momentum, but for small speeds it collapses nicely to $\mathbf{p} = m\mathbf{v}$ as $\gamma \rightarrow 1$ so we don't notice in standard mechanics.

This doesn't resolve the issue with massless photons yet, but does fix the first point!

2 Relativistic Kinetic and Total Energy

By definition, we have:

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}$$

Now, momentum has an extra factor of γ . Suppose we have a massive object which is initially stationary. We apply a force to that object, in order to give it a kinetic energy. This KE is denoted K_A at the point where it reaches speed v_A under the effect of the force. We assume that all work done is entirely transferred to kinetic energy stores.

$$K_A = \int_0^A \mathbf{F} d\mathbf{x} = \int_0^A \frac{d\mathbf{p}}{dt} d\mathbf{x}$$

And we can use the chain rule to say:

$$d\mathbf{x} = \frac{d\mathbf{x}}{dt} dt \implies K_A = \int_0^A \frac{d\mathbf{p}}{dt} \frac{d\mathbf{x}}{dt} dt = \int_0^A \frac{d\mathbf{p}}{dt} v dt \quad (1)$$

We can also say that:

$$\frac{d}{dt}(pv) = p \frac{dv}{dt} + v \frac{dp}{dt}$$

So substituting this into eqn. 1 we have:

$$\begin{aligned} K_A &= \int_0^A \left[\frac{d}{dt}(pv) - p \frac{dv}{dt} \right] dt \\ K_A &= \int_0^A \frac{d}{dt}(pv) dt - \int_0^A p \frac{dv}{dt} dt \\ K_A &= [pv]_0^A - \int_0^{v_A} p dv \\ K_A &= \gamma m v_A \cdot v_A - \int_0^{v_A} \gamma m v dv \\ K_A &= \gamma m v^2 - m \int_0^{v_A} v \left(1 - \frac{v^2}{c^2} \right)^{-1/2} dv \end{aligned}$$

Note that:

$$\frac{d}{dv} \left(\left[1 - \frac{v^2}{c^2} \right]^{+1/2} \right) = \frac{1}{2} \left(1 - \frac{v^2}{c^2} \right)^{-1/2} \left(-\frac{2v}{c^2} \right) = \frac{-v}{c^2} \left(1 - \frac{v^2}{c^2} \right)^{-1/2}$$

Hence:

$$\begin{aligned} K_A &= \gamma m v_A^2 = m \left[-c^2 \left(1 - \frac{v^2}{c^2} \right)^{+1/2} \right]_0^{v_A} \\ K_A &= \gamma m v_A^2 + mc^2 \left(\left[1 - \frac{v_A^2}{c^2} \right]^{+1/2} - 1 \right) \\ K_A &= \gamma m v_A^2 + mc^2 \left(\frac{1}{\gamma} - 1 \right) \end{aligned}$$

And multiplying through by gamma to allow factorisation:

$$\begin{aligned} K_A &= \gamma m v_A^2 + \gamma m c^2 \left(1 - \frac{v_A^2}{c^2} \right) - mc^2 \\ K_A &= \gamma m v_A^2 + \gamma m c^2 - \gamma m v_A^2 - mc^2 \\ \boxed{K_A} &= mc^2 (\gamma - 1) \end{aligned}$$

This gives the kinetic energy of an object with speed v_A with γ as standard. This yields the low energy limit $K = mv^2/2$ as $V_A/c \rightarrow 0$ as expected. Einstein suggested the breakdown of this into two terms:

$$(\gamma - 1)mc^2 = \underbrace{\gamma mc^2}_{\text{total energy}} - \underbrace{mc^2}_{\text{mass energy}}$$

For almost all purposes, and all purposes in this course, we will refer to the total energy and not the overall total energy:

$$E_{\text{tot}} = E\gamma mc^2$$

This is not Lorentz invariant, but is conserved in decays/collisions etc.

3 Forces

By definition we have:

$$\vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(\gamma m v)$$

This can be performed to show that:

$$F = \gamma^3 m a$$

As $\gamma \rightarrow \infty$ as $v \rightarrow c$, it gets harder and harder to accelerate a particle as the particle becomes ultra-relativistic.

4 Lorentz Transformations for Energy and Momentum

There are equivalent transformations for energy and momentum. It turns out than we can use exactly the same Lorentz transformations if we make the following substitutions:

$$[ct, x, y, z] \rightarrow \left[\frac{E}{c}, p_x, p_y, p_z \right]$$

This “grouping three coordinates with another quantity” is broadly called a 4-vector.

Where we previously had:

$$x = \gamma(x' + vt') \quad , \quad t = \gamma\left(t' + \frac{v}{c^2}x'\right)$$

We now substitute $x \rightarrow p_x, ct \rightarrow E/c \implies t \rightarrow E/c^2$:

$$p_x = \gamma\left(p'_x + v\frac{E'}{c^2}\right)$$

$$\frac{E}{c^2} = \gamma\left(\frac{E'}{c^2} + \frac{v}{c^2}p'_x\right) \implies E = \gamma(E' + vp_x)$$

And to transform backwards we use the same logic of replacing all the primed variables with unprimed variables and reverse the sign of the speed. We rarely use this in practice. Instead, we want to find an energy-momentum quantity which is Lorentz Invariant.

5 Energy-Momentum Invariant

Recall that:

- $p = \gamma mv$
- Instead of kinetic energy we use total energy $E = E_{\text{tot}} = \gamma mc^2 = \text{KE} - \text{mass energy}$

We want to continue using $(t, x) \rightarrow \left(\frac{E}{c^2}, p_x\right)$. Using the space-time invariant, we have:

$$c^2t^2 - x^2 - y^2 - z^2$$

Is the energy-momentum invariant simply the space-time invariant but with the energy/momentum transformations?

$$E^2 - (p_x^2 - p_y^2 - p_z^2)c^2$$

This is just a postulate, but does turn out to be a Lorentz invariant. To check its true, we can substitute:

$$\begin{aligned} E^2 - p^2c^2 &= \gamma^2 m^2 c^4 - \gamma^2 m^2 v^2 c^2 \\ &= \gamma^2 m^2 c^2 (c^2 - v^2) = \frac{m^2 c^2 (c^2 - v^2)}{1 - \frac{v^2}{c^2}} = \frac{m^2 c^2 (c^2 - v^2)}{1/c^2 (c^2 - v^2)} \end{aligned}$$

Hence:

$$E_{\text{tot}} - p^2c^2 = m^2c^4$$

As the mass is a constant between frames, as is the speed of light, this is invariant between different frames.

This gives us the invariant:

$$\boxed{E^2 = m^2c^4 + \vec{p}^2c^2}$$

Thu 12 Mar 2026 15:00

Lecture 15 - Special Relativity VI: EM Invariant and CoM Frame

1 Energy-Momentum Invariant

$$E^2 - \vec{p}^2 c^2 = m^2 c^4$$

In the low energy limit, $p \rightarrow c$ and hence $E \rightarrow mc^2$. Here, mass energy dominates.

In the ultra-relativistic limit, $E^2 = p^2 c^2 \rightarrow E = pc$. For a photon, $E = h\nu$ hence:

$$h\nu = pc \implies p = \frac{h\nu}{c} = \frac{h}{\lambda}$$

Which is de Broglie's law.

This quantity is:

- Invariant between all inertial frames.
- Conserved before and after a reaction.
- Applicable to a single particle in which point m is the mass of the particle.
- Also applicable to a system of particles. We define a new quantity:

$$s = \left(\sum E\right)^2 - \left(\sum \vec{p}c\right)^2$$

2 Centre of Mass (Momentum) Frame

We often want to consider a system of multiple particles. The centre of mass frame (CMS frame) is the frame in which the system has no net momentum. If we have two particles in this system colliding head on, they may have different masses and energies but have equal and opposite momenta.

The quantity:

$$s = \left(\sum E\right)^2 - \left(\sum \vec{p}c\right)^2$$

Now reduces to:

$$\sigma E = \sqrt{s}$$

Where $\sqrt{s}$ is called the centre of mass energy. It is the total energy of all particles when evaluated in the CMS frame. Since it's invariant, it's equal to this same number when evaluated in any frame, before or after the collision.

3 Example: LHC

In the lab frame of the LHC, protons of energy of $7\text{TeV} = 7 \times 10^{12}\text{eV}$ collide head on.

3.1 Q1. What is the centre of mass energy $\sqrt{s}$ in this collision?

Here, the lab frame and the centre of mass frame turn out as being the same frame. The total energy therefore is simply $14 \times 10^{12}\text{eV}$

3.2 Q2. What is the maximum mass m_x of the new particle X that could be produced in this collision?

We are working in the lab frame. Initially we have:

$$(\sum E)_{\text{initial}}^{\text{lab}} = 14\text{TeV} \quad , \quad (\sum pc)_{\text{initial}}^{\text{lab}} = 0$$

And finally, assuming that all energy is transferred to particle production:

$$\begin{aligned} (\sum E)_{\text{final}}^{\text{lab}} &= m_x c^2 \\ (\sum pc)_{\text{final}}^{\text{lab}} &= 0 \end{aligned}$$

And by invariance:

$$14\text{TeV} = m_x c^2 \implies m_x = \frac{14\text{TeV}}{c^2}$$

3.3 Q3 Assume we now have a fixed target collider where a moving proton hits a stationary proton in the lab frame. How much energy would this moving proton need to maintain the same CMS frame energy.

As we now have a fixed target collider the laboratory frame appears different. We want to keep the same $\sqrt{s}$ hence the CMS frame is the same.

In the lab frame, we have a proton moving with some energy and momentum $E_p, \vec{p}_p$ colliding with proton 2 which only has mass energy m_p . What E_p do we require to keep $\sqrt{s} = 14\text{TeV}$?

Applying the invariant:

$$s = (\sum E)^2 - (\sum pc)^2$$

The lab frame and the CMS frame are no longer the same. Initially in the lab frame we have:

$$(\sum E_{\text{initial}}^{\text{lab}}) = E_p + m_p c^2 \quad , \quad (\sum pc)_{\text{initial}}^{\text{lab}} = \vec{p}_p c$$

Inserting them into the invariant with the known value:

$$\begin{aligned} (14\text{TeV})^2 &= (E_p + m_p c^2)^2 - p_p^2 c^2 \\ \implies 196\text{TeV}^2 &= E_p^2 + m_p^2 c^4 + 2m_p c^2 E_p - p_p^2 c^2 \end{aligned}$$

And applying the invariant $E_p^2 - p_p^2 c^2 = m_p^2 c^4$ for the single particle:

$$\implies 196\text{TeV}^2 = 2m_p^2 c^4 + 2m_p E_p c^2 = 2m_p c^2 (m_p c^2 + E_p)$$

As we're in an ultra-relativistic limit, we neglect the negligibly small mass energy. This gives:

$$E_p = \frac{196\text{TeV}^2}{2m_p c^2}$$

This gives:

$$E_p = 98 \times \frac{(10^3 \text{GeV})}{\text{GeV}} \approx 100,000\text{TeV}$$

Which is stupidly larger. This is why a colliding beam collider is generally preferable for high energy cases compared to a fixed-target collider.

3.4 Q4

Suppose the particle X given in Q2 turns out to be real. This particle decays into two components:

- A new particle Y with mass $10\text{TeV}/c^2$.
- A massless photon.

What is the momentum of the photon in the lab frame? Initially in the lab frame, which is the CMS frame given the equal and opposite momenta, we only have a stationary particle X with $m_x = 14\text{TeV}/c^2$.

Finally, in the lab(/CMS) frame we have:

- The photon with p_γ, E_γ .
- Particle Y with $p_y, E_y, m_y = 10\text{TeV}/c^2$

By the definition of being in the CMS frame, the total momentum must be zero. Therefore, taking magnitudes:

$$p_Y = p_\gamma = p$$

We apply the invariant:

$$E^2 = p^2 c^2 = m^2 c^4$$

to the photon and the Y -particle. For the photon this gives:

$$E_\gamma^2 - p_\gamma^2 c^2 = 0 \implies E_\gamma = pc \quad (1)$$

Less trivially for the Y -particle, this gives:

$$E_Y^2 - p_Y^2 c^2 = m_Y^2 c^4$$

As the two momenta have equal magnitudes:

$$E_Y^2 - E_\gamma^2 = m_Y^2 c^4 \quad (2)$$

We can now conserve total energy before and after the decay:

$$m_x c^2 = E_y + E_\gamma$$

And eliminating E_Y gives:

$$m_x c^2 = (m_x c^2 - E_\gamma) + E_\gamma$$

Substituting $E_Y = m_x c^2 - E_\gamma$ into (2):

$$(m_x c^2 - E_\gamma)^2 - E_\gamma^2 = m_Y^2 c^4$$

Expanding the left hand side:

$$m_x^2 c^4 - 2m_x c^2 E_\gamma + E_\gamma^2 - E_\gamma^2 = m_Y^2 c^4$$

The E_γ^2 terms cancel:

$$m_x^2 c^4 - 2m_x c^2 E_\gamma = m_Y^2 c^4$$

Rearranging for E_γ :

$$2m_x c^2 E_\gamma = m_x^2 c^4 - m_Y^2 c^4 = (m_x^2 - m_Y^2) c^4$$

$$E_\gamma = \frac{(m_x^2 - m_Y^2) c^4}{2m_x c^2} = \frac{(m_x^2 - m_Y^2) c^2}{2m_x}$$

Since $E_\gamma = pc$ from (1), we divide both sides by c :

$$p = \frac{(m_x^2 - m_Y^2) c}{2m_x}$$

Substituting $m_x = 14\text{TeV}/c^2$ and $m_Y = 10\text{TeV}/c^2$:

$$p = \frac{((14)^2 - (10)^2) \text{TeV}^2/c^4 \cdot c}{2 \times 14 \text{TeV}/c^2} = \frac{(196 - 100)}{28} \text{TeV}/c = \frac{96}{28} \text{TeV}/c$$

$$p = \frac{24}{7} \text{TeV}/c \approx 3.43 \text{TeV}/c$$

Tue 17 Mar 2026 11:00

Lecture 16 - Orbits II: Orbits in Attractive Potentials

Recap from last lecture:

- The magnitude of the angular momentum is conserved: $l = mr^2\dot{\theta}$.
- Total energy is conserved:

$$E = \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} + V(r)$$

This is called the radial energy equation.

1 Interpreting the Radial Energy Equation

For a central conservative potential (fixed energy):

$$E = \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} + V(r)$$

This equation tells us about the behaviour of a particle with constant energy and angular momentum in a radial potential.

The second term looks like a strong repulsive force at a small radius. It effectively flings the particle away if it gets too close. It has the same form as the fictional centrifugal force, so is called centrifugal potential.

We often lump the last two terms together to the “effective potential”:

$$E = \frac{1}{2}m\dot{r}^2 + \underbrace{\frac{l^2}{2mr^2} + V(r)}_{\text{effective potential, } V_{\text{eff}}}$$

Total energy is conserved (so does not rely on r) but the effective potential does. The effective potential gives the minimum possible energy that an object can have with radius of orbit r and angular momentum l . The first term is zero in circular motion (ideal low energy case).

More generally in the non-circular case, the object occupies the space above an ideal sketch for the circular case.

2 Consequences for Attractive Potentials

Note that attractive potentials are negatively signed.

2.1 Example 1: $V(r) = -k/r^3$, where $k \in \mathbb{R}$

In terms of a force:

$$\underline{F}(r) = -\nabla V = -\frac{\partial V}{\partial r} \underline{\hat{r}} = \frac{3k}{r^4} \underline{\hat{r}}$$

The effective potential is given by:

$$V_{\text{eff}} = \frac{l^2}{2mr^2} - \frac{k}{r^3}$$

We want to understand the shape of $V_{\text{eff}} = \frac{l^2}{2mr^2} - \frac{k}{r^3}$ by finding its asymptotes, axis crossings, and turning points in order to sketch it.

Asymptotes

As $r \rightarrow \infty$, $V_{\text{eff}} \rightarrow \frac{l^2}{2mr^2} \rightarrow 0$ (from above).

As $r \rightarrow 0$, $V_{\text{eff}} \rightarrow \frac{-k}{r^3} \rightarrow -\infty$.

Turning Points

$$\frac{\partial V_{\text{eff}}}{\partial r} = -\frac{2l^2}{2mr^3} + \frac{3k}{r^4}$$

This is zero when $r = \frac{3mk}{l^2}$

Assembling these gives:

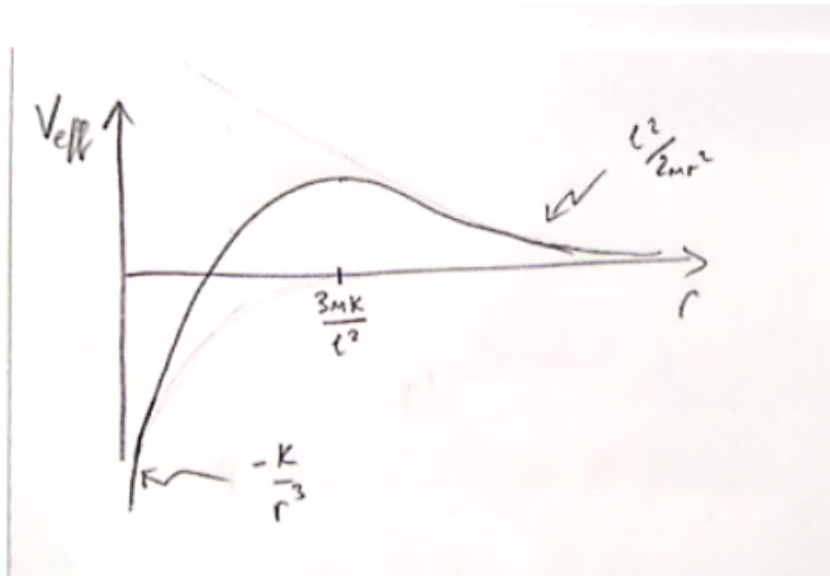


Figure 16.1

As the particle has a combination of potential and kinetic energy, the orbiting particle must have energy lying above this curve. At lowest, when it has no kinetic energy (purely potential) it can lie on this curve, but not below.

There are multiple different particles subject to this effective potential depending on its energy. The maximum value of the potential curve is denoted E_{max} , and the particle has energy E .

- If $E > E_{\text{max}}$.
- If $E < 0$, the particle is always captured and sucked in.
- If $0 < E < E_{\text{max}}$, the particle is either sucked in (down the LHS) or expelled to infinity (along the RHS).

This means that we're dealing with a case where there is no stable possible orbit. This is true for all $V(r) \propto r^{-n}$ where $n \in \mathbb{R}_{\geq 3}$.

2.2 Example 2: $V(r) = -k/r^2$, where $k \in \mathbb{R}$

Depending on the sign of the constants, this gives an effectively $1/r^2$ potential curve. The potential is always positive or always negative, which again results in no stable possible orbits.

2.3 Example 3: $V(r) = -k/r$, where $k \in \mathbb{R}$

$$V(r) = \frac{-k}{r}$$

$$\Rightarrow F(r) = \frac{\partial V}{\partial r} = \frac{k}{r^2}$$

Hence:

$$V_{\text{eff}}(r) = \frac{l^2}{2mr^2} - \frac{k}{r}$$

This time, as:

- $r \rightarrow \infty$: $V_{\text{eff}} \rightarrow -\frac{k}{r} \rightarrow 0$ from below.
- $r \rightarrow 0$: $V_{\text{eff}} \rightarrow \frac{l^2}{2mr^2} \rightarrow +\infty$.

And turning points:

$$\begin{aligned} \frac{\partial V_{\text{eff}}}{\partial r} &= -\frac{2l^2}{2mr^3} + \frac{k}{r^2} \\ &= 0 \text{ when } r_{\min} = \frac{l^2}{mk} \end{aligned}$$

As the curve tends to zero from below at infinity, and to infinity at zero, the turning point must be a minimum. This minimum value at $r = r_{\min}$.

$$\begin{aligned} V_{\text{eff}}(r_{\min}) &= \frac{l^2}{2m} \left(\frac{m^2 k^2}{l^4} \right) - \frac{mk^2}{l^2} \\ V_{\text{eff}}(r_{\min}) &= \frac{mk^2}{2l^2} - \frac{mk^2}{l^2} \end{aligned}$$

And finally the x-axis crossing:

$$\begin{aligned} V_{\text{eff}}(r) &= 0 - \frac{l^2}{2mr^2} - \frac{k}{r} \\ \text{when: } r &= \frac{l^2}{2mk^2} \end{aligned}$$

And finally putting this all together:

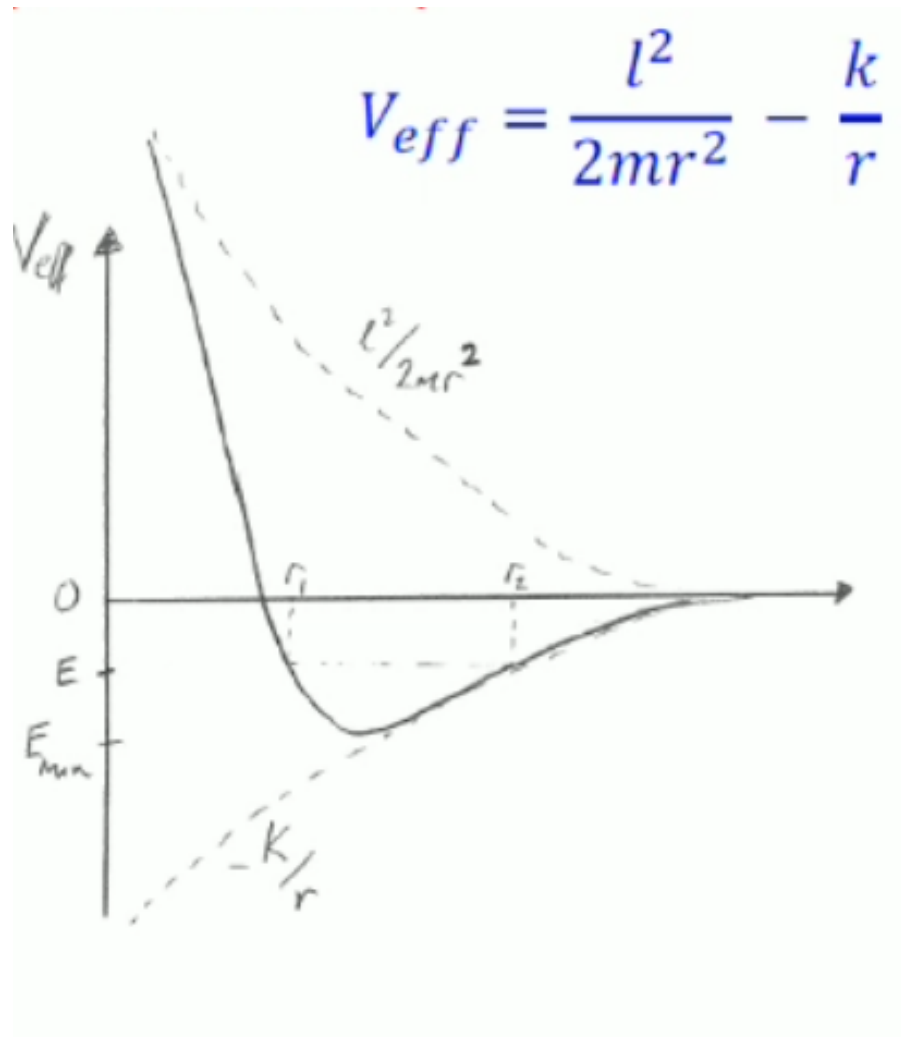


Figure 16.2

This creates a small stable potential well where the orbiting object is bound. Depending on the particle energy, we have a few different cases:

- $E < E_{\min}$: This is not possible. The particle must always have total energy exceeding the minimum potential energy.
- $E = E_{\min} = -\frac{mk^2}{2l^2}$: The particle is in a perfect circular orbit. This is the lowest possible potential state, which (as demonstrated before) requires a fixed $\vec{r}$ circular path.
- $E_{\min} < E < 0$: This is an orbit where the particle is trapped in a stable orbit. The orbit is elliptical, as the particle travels up/down the potential it transfers between kinetic and potential.
 1. In the diagram, with the value of E chosen, the potential is maximum for this E at r_1, r_2 . Here the energy is only potential. For all other values, $E > V_{eff}$.
 2. This gives an elliptical orbit with r_1 and r_2 as the semiminor and semimajor axes.
- When $E = 0$ the particle just barely escapes. It takes a parabolic path.
- When $E > 0$, the particle is certain to escape. It takes a hyperbolic path.

Note that as angular momentum increases, the minimum moves to a larger and larger radius and becomes less and less negative.

3 Parametrising an Ellipse

3.1 Hyperbolic Orbits

When $E > 0$, we have a hyperbolic orbit. This is an unbound object that comes in from infinity, swings around the central object and back out to infinity without being captured.

3.2 Parabolic Orbits

When $E = 0$, we have a parabolic orbit. This is the boundary case where the object just barely escapes. It comes in from infinity, swings around the central object and back out to infinity, but it takes an infinitely long time to do so.

3.3 Bound Elliptical Orbits

When $E < 0$ we have an elliptical orbit (or a circular orbit which is a special case of an ellipse). This is a bound object that is captured by the central object and orbits around it indefinitely.

We can parametrise the ellipse by:

- The shortest distance from the ellipse to the centre, the semi-minor axis b .
- The longest distance from the ellipse to the centre, the semi-major axis a .
- The eccentricity e , which is a measure of how much the ellipse deviates from a circle. It is defined as $e = \frac{a-b}{a+b}$. For a circle, $a = b$ so $e = 0$.

For an ellipse centred on the origin, the equation for an ellipse is given by:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Considering one focus on the ellipse at the origin, we now have:

$$\frac{(x - a\epsilon)^2}{a^2} + \frac{y^2}{b^2} = 1$$

The distance from the focus to the centre is ϵa .

The closest point on the ellipse to the focus is the perigee, at $r = (1 - \epsilon)a$ and the furthest point is the apogee, at $r = (1 + \epsilon)a$.

Tue 24 Mar 2026 11:00

Lecture 17 - Special Relativity VII: Kinematics Examples, Doppler and Natural Units

1 Example I

As viewed in a lab frame, a cosmic ray proton of energy E_p hits a stationary proton. What is the minimum threshold energy of E_p that allows the reaction:

$$p + p \rightarrow p + p + \pi$$

To take place, given $m_p = 938\text{MeV}/c^2$ and $m_\pi = 104\text{MeV}/c^2$?

Let the cosmic ray proton be proton one with E_p, p_p and the stationary proton be proton 2 with m_p .

We can consider energy and momentum in the lab frame, initially (before the collision):

$$(\Sigma E)_{\text{initial}}^{\text{lab}} = E_p + M_p c^2 \quad \text{and} \quad (\Sigma p)_{\text{initial}}^{\text{lab}} = p_p \quad (1)$$

For a threshold energy, the produced particles have no relative motion (i.e. at rest in the centre of mass frame). This means that all available relativistic energy goes into particle production with no excess.

Hence in the centre of mass frame finally, we have two stationary protons and one stationary pion. This is the lowest energy state we can have in the centre of mass frame, with all these particles still existing.

$$(\Sigma E)_{\text{final}}^{\text{CMS}} = (2m_p + m_\pi)c^2 \quad \text{and} \quad (\Sigma p)_{\text{final}}^{\text{CMS}} = 0 \quad (2)$$

We can then apply the energy momentum invariant:

$$s = [(\Sigma E)^2 - (\Sigma p)^2]_{\text{final}}^{\text{lab}} = [(\Sigma E)^2 - (\Sigma p)^2]_{\text{CMS}}^{\text{final}} \quad (3)$$

Inserting 1, 2 into 3:

$$\begin{aligned} (E_p + m_p c^2)^2 - p_p^2 c^2 &= (2m_p + m_\pi)^2 c^4 \\ \Rightarrow E_p^2 + m_p^2 c^4 + m_p c^2 E_p - p_p^2 c^2 &= (2m_p + m_\pi)^2 c^4 \end{aligned}$$

For a single particle, $E_p^2 - p_p^2 c^2 = m_p^2 c^4$, hence:

$$\begin{aligned} 2m_p^2 c^4 + 2m_p c^2 E_p &= (2m_p + m_\pi)^2 c^4 \\ E_p &= \frac{(2m_p + m_\pi)^2 c^4 - 2m_p^2 c^4}{2m_p c^2} \end{aligned}$$

And we have known masses m_p, m_π hence (noting that the masses are in units energy per speed of light squared, so the c s cancel out):

$$E_p = \frac{(2 \times 938 + 104)^2 - 2 \times 938^2}{2 \times 938} = 1228\text{MeV}$$

2 Example II

In an LHC experiment, a B_0 meson ($b\bar{d}$) with mass $5\text{GeV}/c^2$ is produced in a lab frame momentum of $50\text{GeV}/c^2$. The meson has proper decay time $t_0 = 2 \times 10^{-12}\text{s}$. How far does the experiment observe the meson fly before decay?

Let the range of the B_0 in the lab frame be:

$$d_{\text{lab}} = v_{\text{lab}} t_{\text{lab}}$$

There is however a time dilation between the lab frame and the rest frame, hence:

$$t_{\text{lab}} = \gamma t_0$$

$$\implies d_{\text{lab}} = \gamma v_{\text{lab}} t_0$$

To find γv_{lab} note that $p_{\text{lab}} = \gamma m v_{\text{lab}} \implies \gamma v_{\text{lab}} = \frac{p_{\text{lab}}}{m}$. Hence:

$$d_{\text{lab}} = \frac{p_{\text{lab}} t_0}{m}$$

3 Relativistic Longitudinal Doppler Effect

We want to build an analogue for the sonic doppler effect for light. This is especially important in astronomy, i.e. tracking the movement of cosmic spectral lines to determine speeds of distant sources.

We will only consider the longitudinal effect, i.e. the movement of bodies directly towards or away from an observer in their line of sight. We will not (yet) consider transverse effects in this course.

This comes from two components:

- The standard doppler effect (as for sonic waves).
- Relativistic time dilation.

The derivation isn't examinable, but the result for a receding source (redshift) is:

$$f = f_0 \sqrt{\frac{c-v}{c+v}}$$

And for an approaching source:

$$f = f_0 \sqrt{\frac{c+v}{c-v}}$$

4 Problem Solving

1. The energy-momentum invariant is extremely useful. A lot of problems are solved by equating $E^2 - p^2 c^2$ before and after a reaction or a collision.
2. If a problem requires finding γ or $v\gamma$ it's often useful to use $p = \gamma m v$ or $E = \gamma m v^2$ so solve for them instead of manually faffing with u and c .
3. Given a rest frame lifetime t_0 , the range is given by $v\gamma t_0$ or pt_0/m .

Thu 26 Mar 2026 15:00

Lecture 18 - Orbits III: Total Orbital Energy and Perturbed Orbits

1 Eccentricity

We can classify orbits by their eccentricity ϵ :

- $\epsilon = 0$: Circular orbit.
- $0 < \epsilon < 1$: Elliptical orbit.
- $\epsilon = 1$: Parabolic orbit.
- $\epsilon > 1$: Hyperbolic orbit.

Earth orbits very close to a circular orbit, with $\epsilon \approx 0.017$. Bound comets with very large orbits have ϵ close to 1. Unbound comets have $\epsilon > 1$.

2 Total Energy in an Elliptical Orbit w/ $1/r$ Central Potential

Consider a planet with mass m orbiting a star with much larger mass M , in a $1/r$ central potential.

At perigee, the planet has velocity v_1 and away at apogee the planet has velocity v_2 . The $M \gg m$ condition allows us to neglect motion of the star.

We have total energy:

$$E_{\text{tot}} = \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} - \frac{k}{r} = \text{const.}$$

Noting that l is a constant, and $k = GMm$. As the apogee and perigee are turning points, $\dot{r} = 0$ here. Hence:

$$E_{\text{tot}} = \frac{l^2}{mr_1^2} - \frac{k}{r_1} = \frac{l^2}{mr_2^2} - \frac{k}{r_2} \quad (1)$$

$$\frac{l^2}{2m} \left(\frac{1}{r_1^2} - \frac{1}{r_2^2} \right) = k \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$\frac{l^2}{2m} \left(\frac{r_2^2 - r_1^2}{r_1^2 r_2^2} \right) = k \frac{r_2 - r_1}{r_1 r_2}$$

$$l^2 = 2mk \frac{r_2 - r_1}{r_1 r_1} \frac{r_1^2 r_2^2}{r_2^2 - r_1^2}$$

$$l^2 = \frac{2mk(r_2 - r_1)r_1 r_2}{(r_2 + r_1)(r_2 - r_1)}$$

$$l^2 = \frac{2mkr_1 r_2}{r_1 + r_2} \quad (2)$$

Eqn (2) into (1):

$$E_{\text{tot}} = \frac{2mkr_1 r_2}{r_1 + r_2} \frac{1}{2mr_1^2} - \frac{k}{r_1} = \dots \text{for the RHS, but not required.}$$

$$E_{\text{tot}} = \frac{r_2 K}{(r_1 + r_2)r_1} - \frac{k}{r_1}$$

$$\begin{aligned} &= \frac{k}{r_1} \left(\frac{r_2}{r_1 + r_2} - 1 \right) \\ &= -\frac{k}{r_1 + r_2} \end{aligned}$$

This finally gives us:

$$E_{\text{tot}} = \frac{-k}{2a}$$

Where a is the length of the semimajor axis, and $2a$ is the whole length of the major axis.

3 Perturbed Orbits

An impulsive force is applied for a very short time to create an instantaneous change in velocity Δv at some point in the orbit. This changes the total energy of the system, and hence the shape of the orbit.